# MoCHII User Guide

**Abstract**

How to build and run MoCHII (MOnte Carlo for H II regions): requirements, the complete input-parameter reference, output-file formats, worked examples keyed to the regression tests, and the runbook for adding an ion. The physics and validation are documented separately in `docs/MoCHII_physics.pdf`; the atomic-data fitting code in `docs/MoCHII_fitting.pdf`.

**Current atomic-data scope (2026-10-11).** Every metal element carries ionization stages I–VII (77 stages, 66 transitions); a band (`eion_max`) above the ionization potential of the stage-VII ion of an element present is refused (the lowest is Cl VII, 114.2 eV, so the default 13.6–100 eV band runs). Metal photoabsorption is Verner's `phfit2` subshell sum, with the Nahar (2019) R-matrix table for the Ne III ground term. The default `metal_excitation_model = 'chianti'` takes the cooling fits, the *n*-level atoms and the atomic-hydrogen collision files from `data/atomic/excitation/chianti/`: full CHIANTI 11.0.2 atoms for 67 stages and Stout atoms for Cl VII, Ar II, Ca IV (70 atoms, 6629 levels, 70 fits); `'mocassin_v2'` (55 atoms, 571 levels, 49 fits) is for parity runs. Every stage with an atom cools and emits. The items removed on 2026-10-11 (`'extended'`, `additional_line_cooling`, `fe_levels_full`, `metal_inner_shell_absorption`, `high_charge_exchange`, `pe_scale`) are refused with their replacement named. Benchmark values quoted from runs before the restoration of 2026-10-11 (m11 or earlier) are kept until the converged `m13_restoration` runs replace them; the record is `md/ATOMIC_PHYSICS_RESTORATION_RESULTS_2026-10-11.md`.

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# 1. Requirements and build

- Intel oneAPI MPI Fortran (the Makefile takes `mpiifort` when `ifort` is on the path, else `mpiifx`, else GNU `mpif90`; `make F90=mpif90` forces one), HDF5 (default prefix `/data/opt/hdf5_intel`; override with `make HDF5_PREFIX=...`; `make HDF5=0` builds without it, FITS output only), CFITSIO.
- The SEDust dust-emission library, self-contained under `SEDust/` in this tree. Build it once with `cd SEDust/sed && ./build_lib.sh` (→ `SEDust/sed/lib/libsedust.a` and its `.mod` files, which the Makefile links). The SEDust *data* directory is read at run time and resolved automatically relative to the executable (Section 2.10).
- Python 3 with numpy/scipy (h5py, astropy, matplotlib, PyNeb for the test and fitting scripts).

Build:

```
cd MoCHII
(cd SEDust/sed && ./build_lib.sh)   # on a fresh checkout
make -j16       # -> MoCHII.x   (clean + full rebuild, in parallel)
make -j16 MoCHII.x   # rebuild only what an edit reaches
make DEBUG=1    # bounds/traceback build
```

The SEDust library only needs the one-time `build_lib.sh` step; afterward `make` links the in-tree SEDust library. `make` alone cleans and rebuilds everything (the MoCafe default). Module dependencies are tracked: `src/Makefile.dep`, written by `tools/fortran_module_dependencies.py` from the `module`, `use` and `include` statements and regenerated whenever a source changes, makes every object depend on the objects defining the modules it uses and on the generated tables it includes (`src/*.inc`). So `make MoCHII.x` recompiles exactly what an edit reaches — a new component of `params_type` in `define.f90` rebuilds every object that sees `par` — and `-j` orders itself. (Before 2026-09-25 there was no dependency tracking; an incremental build after such an edit linked objects compiled against the old layout.)
Run:

```
mpirun -np 8 ./MoCHII.x input.in
```

MoCHII is MPI-only. The octree, the opacities and the gas state live in MPI-3 shared memory, one copy on each node, and so (since 2026-10-12) do the arrays computed from the reduced transport tallies that every rank would otherwise hold identically: the H, He and metal photoionization and heating rates, the metal stage fractions behind the opacity, the He I $2^3S$ rate $\Phi_3$, and the diffuse-field leaf tables; the ranks of a node fill disjoint leaves of each. The leaf emissivities of the metal lines (`emis_output`) are summed onto rank 0 in chunks, so only rank 0 holds that table. What each rank keeps for itself is what the photons deposit during a transport pass: the energy-integrated path estimators of the photoionization and heating rates (a few values per leaf and absorber) and, only with `jnu_output` or the SEDust emission, the $n_{\rm bin} \times n_{\rm leaf}$ band tally behind `J_nu` — private, so the transport adds to them without locks, then summed over all ranks. The ionization and thermal solves of the leaves are then divided over *all* ranks of the job, cyclically by rank as the SEDust solve is, and the solved leaves are summed back into every node's shared copy (each leaf has one contributor, so the sum is exact), so adding nodes divides the cell work too (since 2026-09-26; before, every node repeated all leaves and the fixed-$T_e$ solve ran on one rank per node). On several nodes with Intel MPI, for example

```
mpirun -hosts node1,node2 -ppn 36 -np 72 ./MoCHII.x input.in
```

from a directory both nodes see: the outputs are written by rank 0 into the working directory, under the base name of `out_file` (its directory part is dropped).

## 2. Input parameters

All input is one `&parameters` namelist assigning fields of `par`. Parameters not listed keep the defaults shown.

**Refused input.** The input is checked before anything is built, and a refused input ends the run with an `ERROR:` line from rank 0 and exit status 1 on every rank, so `mpirun` returns 1 and a script can test it. Refused are: a missing or unreadable input file or one without a `&parameters` group; an unknown or misspelled item, or a value that does not fit its item (the compiler's message is printed with the line it names and the line before it, since Intel Fortran can name the line after the offending one); a string item outside its set of values — every such item is checked whether or not the run reads it (`car_walk`, `conv_crit`, `file_format`, `dust_density_law`, `amr_type`, `ext_geometry`, `slab_faces`, `slab_top_mode`, `slab_bot_mode`, `src_geometry` in every slot, together with the items whose tables below list their values); any of the MoCafe items of §2.13; an item that was removed or renamed, with the replacement named (`setup.f90::refuse_removed_items`: `metal_inner_shell_absorption`, `additional_line_cooling`, `high_charge_exchange`, `fe_levels_full` and `pe_scale` on 2026-10-11, `case_ab`, `diffuse_field` and `line_case` on 2026-09-25), and `metal_excitation_model = 'extended'`; a band (`eion_max`) above the ionization potential of the last carried stage (VII) of an element present (lowest: Cl VII, 114.2 eV); and the combinations the tables below call refused. A missing `amr_file` ends the run the same way (every rank reads it, so every rank reports). The switches compose: `gas_niter = 0` gives a single rates-only pass; `gas_niter > 0` iterates the ionization; `solve_te` (on by default) adds the thermal balance; `use_metals` (on when the input gives a metal abundance) the trace-metal registry; `recombination_continuum = 'transported'` (the default) the explicit diffuse field (which also turns on the He I

and He II excited-cascade channels `hei_diffuse`, `heii_diffuse` unless the input sets them; `heii_diffuse` set with `'on_the_spot'` transports the He III cascade alone); `refine_front` the I-front re-refinement; `ion_add_dust` the dust coupling.

## 2.1 Photons and RNG

| parameter | default | meaning |
|---|---|---|
| `no_photons` | 1e5 | source packets in each iteration (with `add_fuv` and `fuv_sampling='separate'`: the packets of the ionizing part of the band; the FUV part has `no_photons_fuv` of its own) |
| `no_print` | 1e7 | progress-print stride |
| `iseed` | 0 | Mersenne Twister seed (0 = from `/dev/urandom`, or the clock and process id where that cannot be read; a nonzero seed is offset by rank) |
| `launch_sequence` | 'sobol' | packet launch: `'sobol'` (the default since 2026-09-25) = Owen-scrambled Sobol coordinates for the launch of every source packet (energy, source and position, direction; point, multiple, slab and external sources) and, on a second decorrelated stream, of the diffuse packets, indexed by the global photon number (an MPI-count-independent launch set); draws made during transport (dust scattering) stay pseudo-random; prefer `nphotons` = $2^m$. `'random'` = Mersenne Twister, the default before; every input written before the change pins it |
| `source_cdf_tol` | 1e-8 | relative tolerance of the adaptive piecewise-linear representation of a Planck source spectrum from which packet energies are drawn (a tabulated spectrum is represented exactly); must lie in (0, 1); recorded as `CDFRTOL` in the `_rates` header |
| `qmc_seed` | 12345 | scramble key (`launch_sequence='sobol'`); fixed across the gas iterations; change it for independent replicates (their scatter is the RQMC error estimate) |

| parameter | default | meaning |
|---|---|---|
| use_master_slave | F | how the photons of one transport pass are split over the MPI ranks. F = static cyclic: rank $r$ takes indices $r$+1, $r$+1+$n_{\mathrm{proc}}$, …; T = rank 0 becomes the master, serving batches of num_send_at_once consecutive indices to whichever rank asks next; **the master transports no photons**, so $n_{\mathrm{proc}} - 1$ ranks do the work. Applies to the stellar loop, the diffuse LyC loop and (through the same routine) the peel-off imaging pass; the dust-emission infrared transport and the SEDust solve distribute leaves rather than photons and are unaffected. Measured on hii40_bench at 70 ranks (2026-08-02): ∼20% of the ionizing-band pass, and the run-to-run spread of that pass cut from 28% to 1.5%. Under launch_sequence='random' each rank draws from its own stream, so T also selects the realization — statistically equivalent, not numerically comparable; with 'sobol' the two settings transport the same packet set and differ only by summation order |
| num_send_at_once | 10000 | photon indices in one served batch (use_master_slave=T only) |

## 2.2 Grid and dust density

| parameter | default | meaning |
|---|---|---|
| grid_type | 'car' | 'amr' (octree) or 'car' (single-level Cartesian, leaf list in raster order; no tree and no geometry arrays in memory — cell centers and half-widths are computed analytically from the raster index; refine_front unavailable). 'uniform' is accepted as an alias for 'car' |
| car_walk | 'dda' | car-grid traversal: 'dda' (default) = dedicated incremental Amanatides–Woo walks ($t_{\mathrm{max}}/t_{\triangle}$ per axis, comparisons and additions only — the fastest car walk); 'amr' = the octree walks with integer index arithmetic replacing the neighbor table, a verification mode that is bit-identical to the single-level octree (grid_type = 'amr') |
| amr_file | '' | generic AMR leaf file (FITS/HDF5/text): columns x, y, z, level, nH [, metallicity, xHI, ndust]. If empty, a 'car' grid is built from the namelist instead ('amr' requires the file) |
| nx, ny, nz | 1, 1, 11 | 'car'-from-namelist cell counts (cubic cells required: $2x_{\mathrm{max}}/n_x = 2y_{\mathrm{max}}/n_y = 2z_{\mathrm{max}}/n_z$) |

| parameter | default | meaning |
|---|---|---|
| `xmax, ymax, zmax` | 1, 1, 1 | `'car'`-from-namelist box half-extents [`distance_unit`]; box is $[-x_{\max}, x_{\max}]$ etc. |
| `nH_const` | −1 | if $\geq 0$, override every leaf's $n_{\rm H}$ with this uniform value [cm$^{-3}$] (both `'car'` and `'amr'`); $< 0$ keeps the file/column density |
| `density_file` | ” | read the leaf $n_{\rm H}$ from a uniform 3D density cube onto a `'car'` grid (FITS 3D image, or the MoCHII HDF5 image format; NAXIS1/2/3 = $n_x/n_y/n_z$, values $n_{\rm H}$ [cm$^{-3}$]). $n_x/n_y/n_z$ come from the file, the box is `xmax/ymax/zmax`; an alternative to `amr_file`. `reduce_factor`$> 1$ downsamples it (next row); a `rmax/rmin` cut can mask the cube to a sphere |
| `reduce_factor, centering` | 1, 0 | integer downsampling of `density_file` and its anchor |
| `rmax, rmin` | −999, 0 | spherical density cut on leaf centers (both grids): $n_{\rm H} = 0$ for $r > r_{\max}$ (`rmax`$> 0$) and for $r < r_{\min}$ (`rmin`$> 0$; a shell); radii in `distance_unit` |
| `distance_unit` | 'kpc' | length unit of the grid and of every position: 'kpc'/'pc'/'au', or ” for cm; anything else stops the run |
| `xy_periodic` | F | periodic boundaries in $x$ and $y$: a packet leaving a side face re-enters through the opposite one (the $z$ faces stay open). Required by `source_geometry = 'slab'` (§2.4); allows `nx` = 1 on a `'car'` grid built from the namelist (otherwise `nx` $\geq 2$); refused together with `ion_peel` |

| parameter | default | meaning |
|---|---|---|
| `dust_density_law` | 'global_dgr' | leaf gray opacity: `'none'`, `'global_dgr'` ($n_{\rm H} c_{\rm ext}$·DGR), `'from_file'` (ndust column), `'laursen09'` (static xHI column), `'laursen09_live'` (computed $x_{\rm HI}$, refreshed each iteration); any other value is refused (it used to fall through to `'global_dgr'`) |

| parameter | default | meaning |
|---|---|---|
| `cext_dust` | −1 | reference extinction per H [cm$^2$] at `lambda_ref`, which every leaf's gray opacity is $n_{\rm dust}$ times. It is a property of the grains the run carries, so NEGATIVE (the default) takes it from them — from `dust_model`, or from the table `ion_dust_kext` names — before the grid is built, and the run prints the value it took. A positive value is taken as given, for reproducing a published normalization, and is then printed beside the grains' own number with the ratio, with a warning past 1%. A run with `ion_add_dust = .false.` carries no grains in the band and resolves a negative value to 0 |
| `lambda_ref` | 0.55 | reference wavelength [$\mu$m] of `cext_dust`; the band opacity at each packet energy is the gray opacity times $C_{\rm ext}(E)/C_{\rm ext}(\lambda_{\rm ref})$ of the grains |
| `taumax`, `tauhomo` | −999 | optional renormalization of the gray dust opacity at `lambda_ref`, applied once at grid setup: `taumax`$>0$ scales every leaf so the optical depth from the center along $+z$ to the box face is this value, else `tauhomo`$>0$ so the opacity averaged over the dusty volume times the box half-width is; unset keeps `cext_dust` as it is. The run prints both realized values |
| `DGR`, `Z_global`, `Z_ref` | 1e-2, 0.0134, 0.0134 | dust-model scalings |
| `f_ion_dust` | 0.01 | laursen09 survival of dust in ionized gas |
| `f_pah`, `f_ion_pah` | 0, 0 | PAH share of the reference extinction (laursen09_live) and the PAH survival in ionized gas: the PAH grains are weighted by $x_{\rm HI} + f_{\rm ion,pah}\, x_{\rm HII}$ in the `grain_pe` heating always and in the emission with `sed_pah_live`; 0 = PAHs destroyed in ionized gas |

## 2.3 Ionizing/FUV band and the source

| parameter | default | meaning |
|---|---|---|
| `nnu_ion` | 16 | diagnostic ionizing-energy bins (gates use 32). Packet energies are sampled from the source CDF and every cross section is evaluated at the sampled energy, so changing this count changes only the resolution of the saved tallies, never the physical solution |
| `eion_min`, `eion_max` | 13.598, 100 | ionizing band edges [eV] |

| parameter | default | meaning |
|---|---|---|
| `ion_align_edges` | T | align diagnostic bin edges to the ionization thresholds, so no bin straddles an edge; segments are log-uniform and low-priority metal edges are dropped safely when necessary. Affects the tally resolution only, not the transport. Set F for a single log grid |
| `add_fuv` | F or T | FUV extension: extra log bins on [`efuv_min`, `eion_min`] below the unchanged ionizing bins; transports FUV grain heating and grain photoelectric emission, FUV photoionization of low-threshold metals (Ca I 6.11, Mg I 7.65, Fe I 7.90, Si I 8.15, S I 10.36, C I 11.26, Cl I 12.97 eV) and the $2^3$S photoionization rate $\Phi_3$ below 13.6 eV; `luminosity` stays the IONIZING luminosity ($Q_{\rm H}$ preserved, the FUV part rides on top). An input that leaves it out gets T whenever something in the run absorbs below the H I edge: the He I $2^3$S population (`hei_metastable` or `hei_metastable_coupling`, on by default, with `abund_He`$> 0$), a metal with its neutral ionization potential below 13.598 eV (C, S, Mg, Fe, Si, Cl, Ca), or dust (`ion_add_dust`); F otherwise. Setup prints which of the three set it |
| `efuv_min` | 6.0 or 4.78 | FUV lower edge [eV]. Left out of the input, it is 4.78 eV (the edge of $\sigma_3$ as the code evaluates it; the NIST ionization energy of $2^3$S is 4.7677 eV; below every metal edge) in the same cases as `add_fuv`, else 6.0 |

| parameter | default | meaning |
|---|---|---|
| fuv_sampling | 'separate' | how the source packets are shared between the two parts of the band when add_fuv is on. 'separate': two strata, no_photons packets drawn from each source spectrum conditioned on [eion_min, eion_max] and no_photons_fuv drawn from [efuv_min, eion_min); each packet carries its stratum's luminosity over its stratum's count, and a source is chosen within a stratum by its luminosity there (a spectrum starting above eion_min launches no FUV packet; an ISRF preset stops at the H I edge, so a preset-only run has no ionizing stratum and launches only FUV packets). The ionizing solution keeps all no_photons packets. 'proportional': one CDF over the whole band and no_photons packets in all, so the ionizing share is the luminosity share — 45% for a 40 kK blackbody from 4.78 eV, 6% for 20 kK — and the ionizing noise grows by $1/\sqrt{\text{share}}$; kept so runs made with it reproduce bit for bit |
| no_photons_fuv | no_photons/4 | FUV-stratum packets in each transport pass (fuv_sampling='separate'); left out of the input it is a quarter of no_photons, or all of no_photons when no source has ionizing luminosity (an FUV-only field such as an ISRF preset, where the FUV packets are the only packets). With launch_sequence='sobol' the FUV packets are a Sobol prefix of their own stream, so a power of 2 is preferable. An FUV packet crosses more of the grid than an ionizing one (measured 2026-09-25: 1.6× the path segments, 2.0× with dust scattering), so the default adds ∼40–50% to a pass — about what 'proportional' costs at the same no_photons for a 40 kK star |
| nnu_fuv | 8 | FUV bins (16 recommended with metals) |
| tstar | –999 | Planck source temperature [K] |
| ion_spectrum | ” | global spectrum file (column 1 plus one value column); its column units are set by spectrum_type (§2.5) |
| spectrum_type | 'shape_ev' | column units of every spectrum-file slot (ion_spectrum, src_spectrum_file, ext_spectrum): 'shape_ev' (arbitrary, normalized to the scale — legacy) or a physical type 'per_ev'/'per_hz'/'per_ang'/'per_um' (absolute); see §2.5 |

| parameter | default | meaning |
|---|---|---|
| `luminosity` | unset | ionizing BAND luminosity [erg/s] of the (single) source; for a `'shape_ev'`/Planck source unset maps to 1.0 (legacy), for a physical-type file unset means DERIVE it from the file integral; with `nsource`$\geq 2$ it becomes the sum of `src_lum` (or the equal-split total when those are unset) |
| `xs_point`, `ys_point`, `zs_point` | 0 | single-source position (code units, recentered box; `nsource=1`) |

*Multiple internal point sources* (the scalars above are the `nsource=1` case):

| parameter | default | meaning |
|---|---|---|
| `nsource` | 1 | number of internal point sources (0–16); 1 keeps the legacy scalars above, $\geq 2$ uses the `src_*` arrays |
| `src_x`, `src_y`, `src_z` | 0 | positions of sources 1..`nsource` (code units) |
| `src_lum` | unset | ionizing luminosities [erg/s] of the sources; set all or none (none = equal split of `luminosity`); a partial set aborts |
| `src_tstar` | –999 | Planck temperature [K] for each source (multi-source runs only; a single source uses `tstar`) |
| `src_spectrum_file` | ” | one multi-column spectrum file for all sources: column 1 then one value column for each source (units set by `spectrum_type`, §2.5); each source is independently normalized so its ionizing segment carries `src_lum`($i$) (a physical type with `src_lum` unset is derived from that column); overrides `src_tstar`; too few columns abort, extra columns are ignored with a note |

*External isotropic field* (composes with the internal sources; ON when `ext_intensity`$> 0$, an ISRF preset is named, or a physical-type `ext_spectrum` file is given):

| parameter | default | meaning |
|---|---|---|
| `ext_intensity` | −1 | band-integrated mean intensity $J$ [erg s$^{-1}$ cm$^{-2}$ sr$^{-1}$] of the external field; the field is ON when $> 0$ and adds to the internal sources; entering ionizing power $L = \pi J A_{\rm surf}$. With a physical-type `ext_spectrum` file or an ISRF preset it is optional: unset = the absolute $J$ is DERIVED from the file/preset, set = the field is RESCALED to this band $J$ (a note is printed) |

| parameter | default | meaning |
|---|---|---|
| ext_geometry | 'rec' | illuminating surface: 'rec' (box faces, face-area-weighted — correct for a non-cubic box) or 'sph' (bounding sphere of radius rmax about the box center; needs rmax> 0) |
| ext_tstar | –999 | external-field Planck temperature [K] |
| ext_spectrum | " | external-field spectrum: a 2-column file (units set by spectrum_type, §2.5; column 2 is an intensity density $J$) OR a reserved ISRF preset name 'draine'/'habing'/'mathis' (FUV-only, absolute, needs add_fuv); priority ext_spectrum > ext_tstar > the global spectrum |
| ext_scale | 1.0 | dimensionless multiplier on an ISRF preset (ignored when ext_intensity> 0, which rescales the preset to a target band $J$) |
| source_geometry | 'point' | legacy external-only shorthand: 'external', 'external_rec', or 'external_sph' forces nsource=0 with the external field on (needs ext_intensity> 0); 'point' keeps the composable form; 'slab' is the plane-parallel slab illumination below |

## 2.4 Plane-parallel slab illumination

With source_geometry = 'slab' (which requires xy_periodic, §2.2) the box is an *xy-periodic slab*: the medium varies in $z$ (uniform in $x, y$ for a true 1D column with nx = ny = 1, or an arbitrary 3D tile with nx, ny > 1 repeated periodically), and light enters through the top ($+z$) and/or bottom ($-z$) face. Each face is either a collimated *beam* at incidence angle $\theta$ or an *isotropic* (Lambert) field, with its own strength. The single physical normalization is the flux $F_z$ crossing the (horizontal) illuminated face: for a beam $F_z = \mu F_n$ ($\mu = \cos\theta$, $F_n$ the flux normal to the beam), for the isotropic mode $F_z = \pi I$ ($I$ the specific intensity). $F_z$ sets the packet weight through $L = F_z A_{\rm zface}$; every result expressed per unit volume is independent of the tile area. Point-observer peel-off is refused under xy_periodic (setup stops if ion_peel is set; the periodic images have no finite-distance observer); the emergent observable is the boundary intensity $I(\mu)$ (§3.3).

| parameter | default | meaning |
|---|---|---|
| slab_faces | 'top' | which z-face(s) are lit: 'top', 'bottom', or 'both' |
| slab_top_mode, slab_bot_mode | 'beam' | per face: 'beam' (collimated) or 'isotropic' (Lambert) |
| slab_top_theta, slab_bot_theta | 0 | beam incidence angle from the inward normal [deg]; 0 is normal incidence |
| slab_top_phi, slab_bot_phi | 0 | beam azimuth [deg] |
| slab_top_source, slab_bot_source | 0 | source strength of each face (band-integrated): the specific intensity $I$ [erg s$^{-1}$ cm$^{-2}$ sr$^{-1}$] for 'isotropic', or the beam flux $F_n$ [erg s$^{-1}$ cm$^{-2}$] for 'beam' |

| parameter | default | meaning |
|---|---|---|
| `slab_nmu` | 20 | number of $\mu$-bins of the emergent $I(\mu)$ output |

The spectrum is the global one (`tstar` / `ion_spectrum`). Both z-faces always escape (no reflecting boundary). `launch_sequence = 'sobol'` is supported (both modes). With dust scattering (`ion_dust_scatter`) the scattered/reflected component is included in the boundary $I(\mu)$.

## 2.5 Spectrum-file formats

A single parameter, `spectrum_type` (default `'shape_ev'`), fixes the column units of *every* spectrum-file slot (`ion_spectrum`, `src_spectrum_file`, `ext_spectrum`). The type sets the units; the slot sets the meaning: an internal-source file gives a luminosity density $L_X$, the external-field file gives an intensity density $J_X$. Column 1 and the value columns are:

| `spectrum_type` | column 1 | internal (cols 2+) | external (col 2) |
|---|---|---|---|
| `'shape_ev'` | $E$ [eV] ↑ | arbitrary per eV (shape of $L_E$) | arbitrary per eV |
| `'per_ev'` | $E$ [eV] | $L_E$ [erg s$^{-1}$ eV$^{-1}$] | $J_E$ |
| `'per_hz'` | $\nu$ [Hz] | $L_\nu$ [erg s$^{-1}$ Hz$^{-1}$] | $J_\nu$ |
| `'per_ang'` | $\lambda$ [Å] | $L_\lambda$ [erg s$^{-1}$ Å$^{-1}$] | $J_\lambda$ |
| `'per_um'` | $\lambda$ [$\mu$m] | $L_\lambda$ [erg s$^{-1}$ $\mu$m$^{-1}$] | $J_\lambda$ |

The external intensity density $J_X$ carries the same spectral unit as the internal $L_X$ times an extra cm$^{-2}$ sr$^{-1}$ (a mean intensity). All files are plain text with `'#'` comment lines; values outside the tabulated range are zero. `src_spectrum_file` is one multi-column file: column 1 shared by all sources, then one value column for each source (extra columns beyond `nsource` are ignored with a note, too few abort).

**The abscissa is a sample grid, not bin centers.** Column 1 lists the points at which the spectral density is *sampled*: the file describes a continuous function, defined between the points by linear interpolation (in $E$, after the unit conversion), and no bin widths are attached to the rows. Packet energies are drawn from that piecewise-linear density itself, whose cumulative distribution is integrated and inverted analytically, and the band luminosity is its integral; the diagnostic band bins (`nnu_ion`, `nnu_fuv`) are filled separately on a 32-point sub-grid of each bin (zero outside the tabulated range), so the file grid and the band grid are independent and the bins never enter the transport. The spacing is arbitrary — nonuniform grids are fine; place points densely where the spectrum varies rapidly, since only linear variation is captured between adjacent points. Abscissa values must be distinct (no duplicates); a `'shape_ev'` file must be ascending in $E$, while the physical types accept any row order (the table is sorted internally after the unit conversion, so a wavelength table may be given wavelength-ascending).

**Absolute vs. shape normalization.** A physical type (`'per_ev'`/`'per_hz'`/`'per_ang'`/`'per_um'`) is *absolute*: the source luminosity is the tabulated integral itself. If the matching scale (`luminosity` / `src_lum`($i$) / `ext_intensity`) is unset it is DERIVED from the file — each source from its column's ionizing-segment integral, the external field from its band-integrated $J$ — and reported in the log. If a scale is set, the spectrum is RESCALED to it (a note is printed) and

the file then acts only as a shape. With `add_fuv` an absolute spectrum fills the FUV bins from the same table, with no separate normalization. `'shape_ev'` keeps the legacy behavior: the shape is normalized to the scale in every case. A spectrum with no luminosity inside the band aborts, as does rescaling a file whose ionizing-segment integral is zero.

**`'shape_ev'` is per eV, not variable-agnostic.** `'shape_ev'` reads its value columns as the shape of the density per eV $L_E$ on the $E$ [eV] grid of column 1. A table of $F_\lambda$ (or $F_\nu$) values re-gridded onto $E$ under `'shape_ev'` therefore omits the Jacobian and is wrong by a factor $E^{-2}$ (the $\mathrm{d}\lambda/\mathrm{d}E$ conversion). To use a wavelength- or frequency-tabulated shape, give it with its matching physical type and set the scale explicitly — the type conversion then applies the Jacobian and the scale RESCALES the file to your target, so the file acts as a shape with the correct units. For example, a wavelength shape normalized to $10^{49}$ ionizing photons' worth of luminosity:

```
par%spectrum_type = 'per_ang'
par%ion_spectrum  = 'wave_shape.txt'
par%luminosity    = 3.0e38
```

A small `'per_ang'` source file for two point sources ($\lambda$ ascending; the second source is FUV-faint):

```
# lambda[A]   L_lam,1        L_lam,2
   300.0      7.80e34        3.4e33
   500.0      6.05e34        1.2e33
   912.0      4.10e34        0.0
```

## 2.6 Initial state and iteration

| parameter | default | meaning |
|---|---|---|
| `xHI_init` | 1.0 | initial $n_{\mathrm{HI}}/n_{\mathrm{H}}$ (an `xHI` file column wins); an ionized start (1e-3) converges much faster than neutral |
| `xHeI_init`, `xHeII_init` | 1, 0 | initial He fractions |
| `restart_file` | ” | start from a previous run's `<base>_rates` file (HDF5 or FITS): $x_{\mathrm{HI}}$, $x_{\mathrm{HII}}$, $x_{\mathrm{HeI}}$, $x_{\mathrm{HeII}}$, $x_{\mathrm{HeIII}}$, $T_e$, $n_e$, the metal photoionization rates and (with `hei_metastable_coupling`) $\Phi_3$, leaf by leaf, matched by position (each new leaf takes the previous leaf containing its center), so the grid may differ; leaves with no previous counterpart keep the `*_init` values. A file written before 2026-09-26 carries no `x_HII` and no `x_HeIII`; they are then taken as $1-x_{\mathrm{HI}}$ and $\max(0, 1-x_{\mathrm{HeI}}-x_{\mathrm{HeII}})$ with a printed note each (beyond the ionization fronts those values have lost their digits until the first cell solve recovers them). The convergence history restarts, so the `'vol'` test needs eight iterations, unless `restart_iteration_history` is set. Gate `tests/restart/` |

| parameter | default | meaning |
|---|---|---|
| rates_write_interval | 0 | every this many gas iterations, write the iterate to <base>_rates_iterate (the _rates layout; header CONVERGD F, STATESOL F, NITERDON the iteration, FINALDX/FINALDTE that iteration's step), replacing the previous one through a temporary file and a rename, so a run stopped at any point can be resumed with restart_file. Not written at the last iteration or after convergence, where _rates follows. 0 or negative = never |
| restart_iteration_history | F | with restart_file: also take the stopping-test histories the file carries (ConvHist_*), so a resumed run of the same physics continues the 8-iteration window of the 'vol' test instead of refilling it. Off by default: after a change of binary, physics or packet count the old steps say nothing about the new iteration. A file without the histories (written before 2026-10-13) is refused; without restart_file the input is refused |
| abund_He | 0.1 | $n_{\mathrm{He}}/n_{\mathrm{H}}$ |
| gas_niter | 0 | iterations (0 = a single rates-only pass) |
| gas_tol | 1e-3 | convergence tolerance on the ionization state (measure set by conv_crit; with 'vol' it bounds the drift still to come in $x_{\mathrm{HII}}$ and helium) |
| conv_crit | 'cell' | convergence measure: 'cell' (cell maxima; stalls at the front-cell noise floor) or 'vol' (volume-integrated $L_1$; noise-robust, recommended); any other value is refused (it used to run as 'cell'). See §2.7 |
| conv_drift_test | T | with conv_crit = 'vol': a measure on its Monte Carlo floor must also show no net drift of the ionized volume over the last 8 iterations (Student's $t < 2.365$, §2.7); F accepts the floor on $d <$ gas_tol alone, which does not see a front still moving slowly under the noise. Read only with 'vol' (a note says so otherwise) |
| ion_relax | 1.0 | under-relaxation on $x$ ($< 1$ damps front oscillation) |

| parameter | default | meaning |
|---|---|---|
| `recombination_continuum` | 'transported' | what becomes of the ground-state recombination photons (the H I, He I, He II ground continua): 'on_the_spot' absorbed where they are made, the balance uses $\alpha_B$ (Baker & Menzel case B for the continuum; the He I photons that can ionize H are given to H by `hei_on_the_spot`, the He II ones to H and $He^0$ by `heii_on_the_spot`); 'transported' (default since 2026-09-26; 'on_the_spot' before, which the tests/ and results/ inputs that relied on it pin) emitted as diffuse-field packets and followed through the grid, the balance uses $\alpha_A$, and the He I/He III cascade channels follow it when the input leaves them out; the metal ground free–bound continua (channel 6) and the free–free emission inside the band (channel 7, $H^+$, $He^+$, $He^{2+}$ and, with `metal_freefree`, the metal ions) are transported with it and are not transported in the other two modes (their photons leave); 'escaping' lost from an optically thin nebula, $\alpha_A$ and no packets. Replaces `case_ab` and `diffuse_field` (2026-09-25) |
| `lyman_lines` | 'on_the_spot' | what becomes of the Lyman-*line* photons ($n \geq 3 \rightarrow 1$) of H I and He II, which selects the SH95 recombination-line tables: 'on_the_spot' (default) absorbed by the ground state and re-emitted as Balmer lines and Ly$\alpha$ (case B; the photon is converted, not turned into heat), 'escaping' optically thin Lyman lines (case A). The same case sets the H I and He II two-photon continua (the $2s$ share of one Storey & Hummer cascade) and the He III cascade of the diffuse field (`heii_diffuse`: in case A every He II Lyman line to the series limit is emitted, which needs `eion_max` $\geq$ 54.36 eV). Independent of `recombination_continuum`: a nebula whose continuum is transported still traps its Lyman lines. Replaces `line_case` |

| parameter | default | meaning |
|---|---|---|
| `recomb_model` | 'detailed_balance' | H/He recombination coefficients: 'detailed_balance' (default; 'badnell_milne' until 2026-09-26, now refused) takes the ground-level $\alpha_1$ from the Milne relation on the transport cross sections, $\alpha_A$ of He I from the Badnell RR+DR fit ($\alpha_B = \alpha_A - \alpha_1$), and $\alpha_B$ of H I and He II from a zero-density Storey & Hummer *nl*-resolved cascade ($\alpha_A = \alpha_B + \alpha_1$; since 2026-09-26, 1.1% below the earlier Badnell-minus-Milne value for H at $10^4$ K); 'hui_gnedin' the Hui & Gnedin (1997) fits, kept to reproduce earlier gates |
| `require_convergence` | F | if the gas iteration has not converged at `gas_niter`, print a warning; when T, abort before writing output. The convergence status is recorded in the `_rates` header either way (`CONVERGD`, `NITERDON`, `FINALDX`, `FINALDTE`) |

## 2.7 Convergence of the gas iteration

The gas iteration (`gas_niter` sweeps of transport → rate integrals → leaf solve → opacity refill) stops when the state stops changing between sweeps. Each sweep compares a *change measure* against a tolerance:

- ionization only (`te_fixed`, no thermal solve): converged when the $x_{\rm HII}$ change $<$ `gas_tol`;
- coupled (`solve_te`): converged when the $x_{\rm HII}$ change $<$ `gas_tol` *and* the $T_e$ change $<$ `gas_tol_te`.

**Two measures (`conv_crit`).** Both are computed every sweep; `conv_crit` selects which one gates the loop.

**`'cell'`** the cell maxima $\max_{\rm leaf}|\Delta x_{\rm HII}|$ and $\max_{\rm leaf}|\Delta T_e|/T_e$. Simple, but the ionization-front cells jitter with Monte Carlo noise and never settle, so $\max|\Delta x_{\rm HII}|$ *stalls* at the front-cell noise floor ($\sim$2–4$\times10^{-2}$ at $4\times10^7$ packets) and a smaller `gas_tol` is unreachable. This is the historical default (kept so the recorded gates reproduce).

**`'vol'`** the volume-integrated $L_1$ changes

$$d_x = \frac{\sum_{\rm leaf} V\,|\Delta x_{\rm HII}|}{\sum_{\rm leaf} V\,x_{\rm HII}}, \quad d_{\rm He} = \frac{\sum_{\rm leaf} V\,|\Delta x_{\rm HeI}|}{\sum_{\rm leaf} V\,(1-x_{\rm HeI})}, \quad \frac{\sum_{\rm leaf} V\,n_e\,|\Delta T_e|}{\sum_{\rm leaf} V\,n_e\,T_e} < \texttt{gas_tol_te},$$

with the $\rm He^+ \rightarrow He^{2+}$ measure added where $\rm He^{2+}$ is at least 1% of the ionized helium. The ionization measures are tested on the *drift still to come*, not on the last step: with $\rho$ the decay ratio fitted to $\ln d$ over the last 8 iterations, a measure passes when $d\,\rho/(1-\rho) <$ `gas_tol`, or, once it has stopped decaying ($\rho \geq 0.95$), when $d <$ `gas_tol` *and* the signed change of the ionized volume, $s = \sum V\,\Delta x_{\rm HII} / \sum V x_{\rm HII}$ (and its helium counterparts), has no mean over those 8 iterations that stands out of its scatter (Student's $t < 2.365$; `conv_drift_test`, on by default, F drops this condition). That second condition is what tells the Monte Carlo floor, where the leaves fluctuate about their fixed point and their changes cancel in $s$, from a front still moving slowly, which changes the ionized volume with one sign every iteration. No measure passes before 8 iterations, restarts included. A test on the step alone stops

too early when $\rho$ is close to 1: on the HII40 benchmark ($\rho \approx 0.86$) it stopped at iteration 37 with the He I front 0.17 pc short of where it settles. Of the 80-iteration run that showed this (2026-09-25), the drift test of that day stopped at 58, at the 80-iteration answer; the window test described here has not been replayed on it, since that run's log carries no signed change $s$. Reading the floor off the step alone was not enough either: a restart whose front was still moving slowly ($\rho \approx 0.97$) passed after 4 iterations because one noisy step ratio reached 1. Each iteration prints $d$, $\rho$, the drift, the mean signed step $s$ and its $t$ for $x_{\rm HII}$ and helium (`fixed point` lines). Front jitter occupies little volume (a few thousand front cells against $\sim 10^5$ ionized ones), so it enters the norm suppressed by that ratio; the $n_e$ weight drops the vacuum and `te_min`-pinned no-heating cells that otherwise dominate $\max|\Delta T_e|$ in FUV/PDR runs. These measures decay smoothly — the recommended production setting.

**The Monte Carlo floor.** Even the volume measures do not reach zero: they floor at a Monte Carlo level set by the packet count, scaling as $1/\sqrt{N_{\rm packets}}$ ($\sim 3\times10^{-3}$ in $x$ and $\sim 7\times10^{-3}$ in $T_e$ at $8 \times 10^6$ packets, measured on the FUV dusty Strömgren case). Reaching the floor *is* convergence — iterating further leaves the solution unchanged. So **set `gas_tol`/`gas_tol_te` just above the floor for the packet count in use**: a tolerance below the floor can never be met and the loop runs to `gas_niter` and warns. Rules of thumb (`conv_crit='vol'`): $\sim 5 \times 10^{-3}/10^{-2}$ at $\sim 10^7$ packets, loosened to $\sim 10^{-2}/2 \times 10^{-2}$ at $\sim 10^6$ (the reduced-count `examples/` use the looser values for this reason).

**When it does not converge.** Hitting `gas_niter` without meeting the test is not fatal: the state is still written and a warning is printed on rank 0. Set `require_convergence = T` to abort before the output instead. Either way the `_rates` header records the outcome in `CONVERGD` (0/1), `NITERDON` (sweeps run), `FINALDX`, and `FINALDTE`. The derivation and the gate measurements are in `docs/MoCHII_physics.pdf`.

## 2.8 Thermal balance and metals

| parameter | default | meaning |
|---|---|---|
| `solve_te` | T | bisection on $\log T_e$ in [`te_min`, `te_max`] coupled with the ionization (heating = cooling in every leaf); F holds every leaf at `te_fixed` |
| `te_fixed` | 1e4 | $T_e$ [K] of every leaf when `solve_te` is off; with it on, only the starting $T_e$ of the first pass |
| `te_min`, `te_max` | 1e3, 5e4 | bisection bracket [K] |
| `gas_tol_te` | 1e-3 | convergence tolerance on $T_e$ (with `solve_te`): $\max\|\Delta T_e\|/T_e$ under `conv_crit = 'cell'`, the $n_e$-weighted volume measure under `'vol'` (§2.7) |

| parameter | default | meaning |
|---|---|---|
| `metal_excitation_model` | 'chianti' | the metal atoms (`data/atomic/excitation/<model>/`: levels, radiative and collision rows, the matching cooling fits, the atomic-H collision files), used for the cooling, the level populations, the lines and the charge-exchange product populations. 'chianti' (default): full CHIANTI 11.0.2 atoms for 67 stages and Stout atoms for Cl VII, Ar II, Ca IV (70 atoms, 6629 levels); no line cooling for C VII, Ar I, Cl I, Ca I, Fe I, Cl V, Ca III (no data). 'mocassin_v2': the MOCASSIN v2 atoms (55) plus the CHIANTI Fe II forbidden rows, for parity runs. Every stage with an atom cools and emits. Measured cost of 'chianti' against 'mocassin_v2' (262144 leaves, 11 metals, 72 ranks, 2026-10-11): setup +0.5 min, one iteration +0.45 min, line output +2.7 min, 672 against 647 MB on each rank. 'extended' (removed 2026-10-11) is refused, as are `fe_levels_full` and `additional_line_cooling` |
| `neutral_hydrogen_excitation` | T | atomic-hydrogen impact on the C I, C II, O I ground-term fine structure (Abrahamsson et al. 2007, Barinovs et al. 2005; `hcoll_<el>_<stage>.txt` beside the atom) in the same level equilibrium as the electrons, for cooling and lines; the dominant excitation of these coolants in neutral gas. F only for parity with electron-only codes |
| `metal_collisional_ionization` | T | electron-impact ionization of the metal stages (Voronov 1997, or with `ci_model`) in the stage balance and its ionization-energy loss in the thermal balance. F only for parity with a code that omits it (MOCASSIN v2 reads the data and sets the rate to zero) |
| `langevin_charge_exchange` | F | $X^q + H \rightarrow X^{q-1} + H^+$ for $q \geq 5$ (22 reactions with all eleven elements) at the Langevin capture rate with unit transfer probability, an upper limit, not a measured rate; kinetic heat $2.86(q-1)$ eV. Renamed from `high_charge_exchange` (2026-10-11), which is refused |
| `cooling_model` | 'local_ne' | line cooling in the thermal loop: 'local_ne' (default) suppresses each ion's cooling fit to the local $n_e$ with a $(T, n_e)$ table built from its $n$-level solve, so the thermal balance and the emergent lines are one physics; 'low_density' uses the $n_e \rightarrow 0$ fits directly (the coronal limit, exact in diffuse gas; reproduces earlier gates) |

| parameter | default | meaning |
|---|---|---|
| twophoton_quenching | 'colliders' | collisional $2s \rightarrow 2p$ suppression of the H I two-photon continuum: 'colliders' (default) gives electrons, protons, $He^+$ and $He^{2+}$ each their own Pengelly & Seaton (1964) rate, normalized to Osterbrock & Ferland Table 4.10, with $A_{2s}$ = 8.2246 $s^{-1}$ (Drake 1986; data/atomic/h_2s_2p_mixing.txt); 'lumped_ne' is the MOCASSIN form $1 + n_e q/8.23$ (reproduces earlier gates). Affects _nebcont.txt only |
| hei_on_the_spot | T | with recombination_continuum = 'on_the_spot' only: the $He^+ \rightarrow He^0$ recombination photons that can ionize H are absorbed on the spot — the excited cascade (19.82 eV, 584 Å with the $p_{584}$ split, $2^1S$ two-photon) by $H^0$, the ground captures shared between $H^0$ and $He^0$ by opacity — with $E - 13.598$ eV of heat each (Osterbrock & Ferland eqs. 2.24–2.25). .false. is the earlier case B, which gave every He recombination photon back to helium |
| heii_on_the_spot | T | with recombination_continuum = 'on_the_spot' only: the $He^{2+} \rightarrow He^+$ recombination photons are absorbed on the spot (Osterbrock & Ferland sec. 2.5). The ground captures ($\geq$ 54.416 eV) are shared by opacity among $He^+$, $He^0$ and $H^0$ over the Milne spectrum; $He^+$'s share re-ionizes it (case B), the shares $y_H$, $y_{He}$ ionize $H^0$ and $He^0$, so He III recombines at $\alpha_B + (y_H + y_{He})\alpha_1$. The excited cascade — free–bound into $n \geq 2$, 40.813 eV Ly$\alpha$, the $2^2S$ pair, counted from the same branch luminosities as heii_diffuse — ionizes $H^0$ below 24.587 eV and is shared by $H^0$ and $He^0$ above it. Heat $E - \mathrm{IP}$ of the absorber per photon ($54.418 - \mathrm{IP}$ for a ground capture). Independent of eion_max: the photons are made by the gas, not by the band. $H^0$'s share of the cascade enters hydrogen's balance as an absolute source; a cell can absorb on the spot only as many as the recombinations that make $H^0$, and in a fully $He^{2+}$ zone the cascade outnumbers them for He/H above 0.12–0.15, where no local balance exists (the heat is then limited to what the supply can absorb, the H balance residual shows the leaf, and setup prints a NOTE for abund_He $> 0.12$): use 'transported' there. .false. is the earlier case B, in which $He^{2+}$ recombined at $\alpha_B$ and its photons ionized nothing |

| parameter | default | meaning |
|---|---|---|
| data_dir | ” (auto) | root of the data tables (HeI2phot.dat, gauntff_vh14.dat, the nebular-continuum tables); blank resolves to <executable dir>/data, so a run works from any working directory; a path set here is taken as given |
| atomic_dir | ” (auto) | atomic coefficient files; blank resolves to <data_dir>/atomic |
| use_metals | (F) | registry cascade + metal line cooling. Left out of the input it is set from the abundances the input gives: T if any abund_<El> it sets is $> 0$, F otherwise; metal_ne and metal_heat, when left out, follow it. A value the input sets is kept |
| ci_model | ’voronov’ | collisional ionization: ’voronov’ (default) or ’dere_hybrid’ (Voronov $\times$ the Dere 2007/Voronov ratio; matters only in shielded low-ionization zones) |
| abund_C/N/O/Ne/S | 0 | $n(\mathrm{X})/n(\mathrm{H})$; an element is active when $> 0$. Every metal abundance defaults to 0 (since 2026-09-25), so an input names the metals it carries; the inputs written before then pin the HII20 values they relied on (C $2.2 \times 10^{-4}$, N $4 \times 10^{-5}$, O $3.3 \times 10^{-4}$, Ne $5 \times 10^{-5}$, S $9 \times 10^{-6}$) |
| abund_Ar/Mg/Fe | 0 | further registry elements (gas-phase values; Mg/Fe are heavily depleted) |
| abund_Si/Cl/Ca | 0 | further registry elements (gas-phase; Si and especially Ca heavily depleted, Cl nearly undepleted — Orion-like 3e-6 / 3e-7 / 2e-8); adds [Si II] 34.8 $\mu$m, Si III/IV, [Cl II/III/IV] (5518/5538 density pair), Ca II, [Ca V] among others. Every element carries stages I–VII (77 stages, 66 transitions); a band above the ionization potential of its stage-VII ion is refused (Cl VII 114.2 eV is the lowest) |
| ion_metal_abs | T | metal photoionization absorption in the band opacity (active with use_metals); the only gas opacity in the FUV bins |

| parameter | default | meaning |
|---|---|---|
| `charge_exchange` | 'two_way' | charge exchange $X^{i+} + H^+ \leftrightarrow X^{(i+1)+} + H^0$ in hydrogen's balance as well as the cascade's, so the charge each reaction moves is conserved in the reacting pair: $n_{HI}(\Gamma_H + n_e C_H + R_{cx}^{out}) = (\alpha_H n_e + R_{cx}^{in}) n_{HII}$ with the sums over the active reactions (45 with all eleven elements: 7 ionization, 38 recombination; `langevin_charge_exchange` adds 22). 'metal_only' keeps them in the cascade alone (the earlier bookkeeping, which does not conserve charge); it is a reproduction value, not a physics option. $N^{2+}$ and $O^{2+}$ + H use the Barragan et al. (2006) tables, $N^{4+}$ and $Si^{2+}$ + H the updated KF96 database distributed with CLOUDY (Ferland et al. 2013), the other higher stages KF96 over their published intervals |
| `helium_charge_exchange` | T | He–H charge exchange in the H/He balance: $He^+$ + H $\rightarrow$ He + $H^+$ as the sum of a radiative channel (Stancil, Lepp & Dalgarno 1998 reaction 19, the Zygelman et al. 1989 calculation times the singlet approach probability 1/4; Glover & Jappsen 2007 R26, 4% higher, until 2026-09-26) and a non-radiative one (Loreau et al. 2014 by microscopic reversibility, tabulated by `tools/fitting/make_heii_h_charge_exchange.py`; the Kingdon & Ferland 1996 fit, 93$\times$ higher at $10^4$ K, until 2026-09-26); $He^{2+}$ + H $\rightarrow$ $He^+$ + $H^+$ (Kingdon & Ferland 1996); and the two reverses He + $H^+$ $\rightarrow$ $He^+$ + H (of the non-radiative channel alone) and $He^+$ + $H^+$ $\rightarrow$ $He^{2+}$ + H by detailed balance. Each reaction enters the H and the He balance, so charge is conserved in the pair. Needs nothing else (independent of `use_metals`). F drops all four; it reproduces runs made before 2026-09-24 and is not a physics option |

| parameter | default | meaning |
|---|---|---|
| charge_exchange_heating | T | the kinetic energy the charge-exchange reactions give the gas, $\Delta E$ = [IP(acceptor's product) − IP(donor)] − $E_{\rm rad}$, $E_{\rm rad}$ the product excitation that radiates (for radiative charge transfer the photon; the Coulomb energy of products that repel, several eV for $He^{2+}$ + H, is left out, an approximation marked in ion_balance_mod); negative for endothermic reactions. Mg + $H^+$ is branched in temperature between $Mg^+$ 3p and the endothermic 4s by Allan et al. (1988) (CXHEATTAB), carried below the table's 5000 K by the 4s threshold Boltzmann factor. Added to the thermal balance for the metal–H reactions under use_metals and for the He–H reactions under helium_charge_exchange. The defects are the CXHEAT rows of element_<el>.txt (make_element_data.py, which names each product state and its source); an element file without them is refused while this is T. At the converged Lexington states it was $3.6\times10^{-4}$ (HII40) and $3.0\times10^{-6}$ (HII20) of the photoheating, mostly $O^{2+}$ + H (measured 2026-09-25). The $q = 3, 4$ reactions added with the stages of 2026-10-11 take the Kingdon & Ferland (1999) mean energies. F reproduces the thermal balance of runs made before 2026-09-25 |

| parameter | default | meaning |
|---|---|---|
| jnu_output | F | write J_nu (band mean intensity per bin and leaf) into _rates. The tally behind it is $n_{\rm bin} \times n_{\rm leaf}$ values on *every* MPI rank, and nothing else reads it except the SEDust emission, so it is kept only with this item or dust_sed (off by default since 2026-09-26; the tally was always kept before). The rates never use it |
| emis_output | F | write _emis (HDF5/FITS): leaf-by-leaf line emissivities + the state needed for maps (Section 3.) |
| emis_line_floor | 1e-4 | the metal blocks of _emis carry the lines with $L \geq$ this times $L(\mathrm{H}\beta)$ (a full CHIANTI atom has up to $\sim 10^4$ lines); _lines.txt carries every line |

| parameter | default | meaning |
|---|---|---|
| `hei_metastable` | F | write `_heimeta` (HDF5/FITS): the He I $2^3$S metastable density $n_3$, its photoionization rate $\Phi_3$ and heating $\Gamma_3$, the recombination coefficient into $2^3$S $\alpha_3$, plus $n_\mathrm{H}$, $n_e$, $T_e$, and leaf geometry (the 10833 Å absorption diagnostic; Section 3.). An input that leaves `add_fuv` and `efuv_min` out transports the FUV extension from 4.78 eV, so the soft-UV part of $\Phi_3$ is included |
| `hei_metastable_coupling` | T | couple the He I $2^3$S population (the same $n_3$ `hei_metastable` writes; Bray et al. 2000 electron-impact rates, $\alpha_3 = f_{3S}\alpha_B(\mathrm{He\,II})$) into the ionization balance ($2^3$S ionization by electrons and photons returns He to $He^+$; Penning + associative ionization by $H^0$, García Muñoz 2025, ionizes H), the thermal balance (the excitations into and out of $2^3$S, its ionization, its photoionization heating and Penning, with a closed energy ledger), and, with `hei_diffuse`, the He I diffuse channel (19.82 eV photons = $n_3 A_{31}$; the collisional transfers feed the $2^1$S two-photon and 584 Å branches). $\Phi_3$ covers the transported band only: without `add_fuv` the $[4.78, 13.6)$ eV part is missing, which for a 40 kK star is $\sim 10\times$ the part above 13.6 eV — hence the `add_fuv`/`efuv_min` defaults above. F drops the coupling (the diagnostic `hei_metastable` uses the same population either way) and reproduces the earlier solve bit for bit; it was F between 2026-10-09 and 2026-10-11 |
| `h_coll_effects` | F | add the collisional H I Balmer component (excitation of neutral H) as separate `hc` rows/blocks; matters at fronts and hot cells |
| `diffuse_energy_model` | 'milne' | spectrum of the three ground continua (needs `recombination_continuum = 'transported'`): 'milne' (default) is the detailed-balance free–bound spectrum $E^2\sigma_\mathrm{pi}(E)e^{-(E-E_\mathrm{th})/kT}$ with the transport cross sections; 'exponential' the historical $E_\mathrm{th} + kT[-\ln u]$ approximation (reproduces earlier runs); 'threshold' a sensitivity bracket, not a model |

| parameter | default | meaning |
|---|---|---|
| `hei_diffuse` | on if transported | fourth diffuse channel (needs `recombination_continuum = 'transported'`; left out of the input it is on then, since nothing is absorbed on the spot and these photons would otherwise be lost): He I excited-level recombination photons. The $2^3S : 2^1S : 2^1P$ branch fractions come from one consistent set of $\alpha_{\rm eff}$ (Benjamin, Skillman & Smits 1999 Table 1 with J. S. Mathis, as in K. Wood's `ionize`) normalized by its own sum — 0.762 : 0.076 : 0.162 at 8000 K against the statistical 0.750 : 0.083 : 0.167 — and the $2^1S$ photons from the Drake et al. (1969) two-photon shape. Against the statistical weights with a flat two-photon spectrum this is +0.12% in H-ionizing energy and +0.30% in photons for each case-B He II recombination at 8000 K; in transport (2026-07-30) the branch fractions moved $L(\mathrm{H}\beta)$ at HII40 by +0.012%. The trapped $2^1P$ (584 Å) photon is split locally between photoionizing hydrogen and conversion to $2^1S$ (the 2.06 $\mu$m $2^1P \rightarrow 2^1S$ decay that ends its resonance walk), with $p_{584} = [1 + 77\,(y_{\rm He}/0.1)\,x(\mathrm{He}^0)/(\sqrt{T}\,x(\mathrm{H}^0))]^{-1}$ (their 77 assumes He/H = 0.1) (Wood, Mathis & Ercolano 2004) — the same split the He I two-photon nebular continuum uses. Measured when the split was added (2026-07-30, against the same channel without it), it lowered $L(\mathrm{H}\beta)$ at HII40 by 0.2% and moved the He I front inward by 0.0036 pc |

| parameter | default | meaning |
|---|---|---|
| `heii_diffuse` | on if transported | fifth diffuse channel: what a He III recombination radiates on its way down through He II, in the case of the Lyman lines (`lyman_lines`). With `recombination_continuum = 'transported'` it joins the other channels (left out of the input it is on then when `eion_max` $\geq$ 40.813 eV, and left off with a note below that, where the band cannot make He III). With `'on_the_spot'` it is off unless the input sets it; set, the cascade alone is transported and `heii_on_the_spot` keeps only the $He^{2+}$ ground captures — for a He III zone optically thin at 40.8 eV, where absorbing the cascade in place deposits its $E-\mathrm{IP}$ (up to $\sim$ 27 eV a photon) where the photons in fact leave (a NOTE each iteration counts such leaves). No effect with `'escaping'`. Unlike `hei_diffuse` this is not a small correction — the He II $n = 2$ threshold at 13.604 eV sits just above the hydrogen edge, so the whole Balmer continuum ionizes H, and the $2^2$S pair spans 40.813 eV. Three paths: free–bound into $n \geq 2$ (7.5–10.3% of the channel over $10^4$–$3\times10^4$ K, from the same fb+ff tables the written continuum uses, minus free–free), $2^2$P Ly$\alpha$ at 40.813 eV (64.1–70.0%), and the $2^2$S two-photon continuum (22.5–25.6%, 36.090 eV of the pair landing in band); case B, from `tests/heii_cascade/`. The $n = 2$ exit branching $f_{2s} = \alpha_{\mathrm{eff}}(2^2\mathrm{S})/\alpha_B$ takes both parts from one zero-density Storey & Hummer cascade (`hydrogenic_recombination_mod`) so the ratio closes one calculation; it reproduces MOCASSIN's independent He II Ly$\alpha$ fit to 2.3–6.2%. Matters only where $He^{2+}$ exists: on a $10^5$ K blackbody in the HII40 geometry it raised the diffuse-field luminosity 44% and $L(\mathrm{H}\beta)$ 0.98% (measured 2026-08-09, with the SH95 branching of then); on both Lexington benchmarks, which have no $He^{2+}$, it changes nothing |
| `gaunt_vh14` | F | free–free Gaunt factors from the van Hoof et al. (2014) table (`data/gauntff_vh14.dat`) instead of Hummer (1988); the two agree to $\lesssim$0.5% in $T_e$ |

| parameter | default | meaning |
|---|---|---|
| `metal_ne` | = `use_metals` | electrons from the metal cascade join the $n_e$ closure. Solved as a fixed point, since the stage fractions depend on $n_e$ themselves: a single evaluation at the H/He-only electron density would be the $n_e \to 0$ limit, in which nothing recombines and every low-IP metal comes out fully ionized. Switching it off cost only 0.002% of $\langle T[N_\mathrm{p}N_\mathrm{e}]\rangle$ on the Lexington benchmarks (measured 2026-07-31) — but beyond the ionization front the metal stages are the ONLY electron source, so there it is not optional |
| `metal_heat` | = `use_metals` | metal photoheating in the thermal balance, accumulated with the same Γ's the cascade uses, $\sum_s \sigma_s(1 - E_{\mathrm{th},s}/E)$ over the absorbing subshells of `phfit2` (an inner-shell hole is assumed to leave by fluorescence and one stage is gained; the Auger second electron of Mg I $2p$, Ca I $3p$ and Fe I–III $3p$ is not carried, a known defect). The metal recombination energy losses and, if enabled, collisional-ionization losses (`species_mod::metal_ionization_cooling`) are under the same switch. Switching it off lowered $\langle T[N_\mathrm{p}N_\mathrm{e}]\rangle$ by 21.1 K at HII40 and 5.7 K at HII20 (measured 2026-07-31, before the metal recombination and collisional-ionization losses joined this switch on 2026-09-24) |
| `grain_pe` | with dust | grain charge distribution, photoelectric heating (with photodetachment) and the cooling by captured electrons and ions (Weingartner & Draine 2001) for the grains of `dust_model`, scaled to the leaf's dust opacity, in the leaf's band field; PAH grains weighted by $x_\mathrm{HI}$+ `f_ion_pah`$x_\mathrm{HII}$; astrodust takes the WD01 silicate photoemission parameters (no published astrodust yield). Left out of the input, on when `ion_add_dust` and `add_fuv` are; set without either, refused. It keeps the band tally behind `J_nu`. Replaces the Bakes & Tielens (1994) fits and their `pe_scale`, which is refused. Gate `tests/grain_photoelectric/` |
| `gas_grain_energy_exchange` | with dust | energy neutral H and He exchange with the grains they strike, $\propto \alpha\, 2k(T - T_d)$ (Hollenbach & McKee 1979 accommodation; $T_d$ the equilibrium mixture temperature); left out of the input, on with `ion_add_dust`; set without it, refused |

| parameter | default | meaning |
|---|---|---|
| use_sec_ion | F | secondary ionization by fast photoelectrons (photoelectron_degradation_mod): a photoelectron with $E_0 = h\nu - E_{\rm th} > 30$ eV from H I, He I or He II deposits $f_{\rm heat}(x, E_0)$ as heat (Dalgarno, Yan & Liu 1999 eq. 14 with a $(1-x)$ closure) and spends $f_{\rm ion}(x, E_0)$ ionizing H I/He I (Shull & van Steenberg 1985 amplitudes with DYL99's $E_0$ dependence), divided over the neutrals present; $x = n_e/(n_{\rm H} + n_{\rm He})$ of the current cell-solve iterate; below 30 eV all heat, metal photoelectrons all heat; $\sim$0 in fully ionized gas, matters at fronts and in hard fields |
| metal_freefree | T | include the trace metals in the free–free cooling with the net ion charge $Z_{\rm ion}$ = (stage $-1$), not the nuclear $Z$; a trace ($\sim 10^{-4}$ of the H/He free–free) correctness term, on by default |

## 2.9 I-front re-refinement

| parameter | default | meaning |
|---|---|---|
| refine_front | F | rebuild the octree on the I-front once |
| refine_iter | 20 | at which iteration |
| refine_lbase, refine_lmax | 5, 7 | base/front levels |
| refine_eps | 1e-2 | front flag: $\epsilon < x_{\rm HI} < 1-\epsilon$ |
| refine_dx | 0.2 | or a face-neighbor $x_{\rm HI}$ jump above this |

## 2.10 Dust in the band and dust emission

| parameter | default | meaning |
|---|---|---|
| ion_add_dust | F | dust absorption in the band (dusty Strömgren) |

| parameter | default | meaning |
|---|---|---|
| ion_dust_kext | ” | optional kext table (e.g. data/kext_albedo_WD_MW_3.1_60_D03.all_2003). Blank (the default) takes $C_{\rm ext}$, albedo and $\langle\cos\rangle$ from the grain model named by dust_model, so the transport and the emission refer to the same grains; a file OVERRIDES the model and is then consistent with the emission only if it came from the same grains. Taken from the model, the curve is /kext of that model's own optics product, SEDust/data/<model>/sedust_<model>.h5, which carries the wavelength axis and the cross-section tables in the same file, interpolated onto the model's own grid (sed_kext replaces it). The model must cover the band, and all four do on their own: astrodust spans 1762 wavelengths over $1.0\times10^{-4}$–$3.981\times10^{4}$ $\mu$m from the T-matrix optics table, so below the Lyman limit the optics are the $b/a$ = 1.4 oblate spheroid itself rather than a sphere approximation; DL07 and MRN share their own axis, 1823 wavelengths from $6.2\times10^{-5}$ $\mu$m; and Zubko's DustEM optics span 0.001–$10^{4}$ $\mu$m. Nothing is extended at any energy the code transports; an extension can only be asked for above eion_max = 12398.4 eV, and astrodust refuses it there with a message naming eion_max rather than substitute an equal-volume sphere |
| ion_dust_scatter | F | sample dust scattering (HG, albedo weights) |
| dust_sed | F | SEDust stochastic dust SED at output time (with ion_add_dust) |
| dust_model | 'astrodust' | 'astrodust'/'dl07'/'mrn'/'zubko'. 'mrn' is the Mathis, Rumpl & Nordsieck (1977) graphite + silicate $a^{-3.5}$ power law over 0.005–0.25 $\mu$m on the Draine & Lee (1984) abundances, Mie on the D03 dielectric functions; it has NO PAH component, so a run on it carries no aromatic features and sed_pah_live has nothing to weight |

| parameter | default | meaning |
|---|---|---|
| `sed_data_dir` | ” (auto) | the ONE directory everything the named `dust_model` is made of is resolved from: its optics product `<model>/sedust_<model>.h5`, the size distribution under `release/`, the dielectric functions and PAH cross sections under `dielectric/`, and Zubko's ZDA configuration. Blank auto-resolves to `<executable dir>/SEDust/data`, so a fresh checkout runs the dust path from any working directory |
| `sed_dl07_sdindex`, `sed_dl07_uisrf` | 7, 1.0 | DL07 only: WD01 size-distribution index (7 = MW $R_V = 3.1$, $b_C = 6\times10^{-5}$) and the ISRF intensity setting the PAH ionization fraction. Leave them at the defaults unless you also supply a matching `sed_kext`: the served extinction curve was computed from the defaults, so moving these moves the grains that emit without moving the curve that extinguishes. The gate `tests/grain_kext` checks that pairing |
| `sed_kext` | ” | the precomputed size-integrated extinction curve SEDust serves to the transport ($\lambda$, albedo, $\langle\cos\rangle$, $C_{\rm ext}$/H, $C_{\rm abs}$/H, $C_{\rm sca}$/H), resolved under `sed_data_dir`. Blank (the default) takes `/kext` of the named `dust_model`'s own product, so the curve and the cross sections come from one file and cannot be paired wrongly. Naming a file makes it MANDATORY (the build stops if it cannot be read) and it must have been computed from the same model and size distribution the emission uses, which is what SEDust's `calc_kext.x` writes. Distinct from `ion_dust_kext`, which does not change what SEDust serves but REPLACES the served curve downstream, leaving the emission on other grains |
| `sed_NT`, `sed_Tlo`, `sed_Thi` | 200, 2.7, 5e3 | SEDust temperature grid |
| `dust_emis_bands` | unset | up to 8 wavelengths [$\mu$m] (e.g. 8, 24, 70, 160): SEDust band emissivities of each leaf (with `dust_sed`) → `_dustemis` (blocks `emis_dust/wl_dust` [erg s$^{-1}$ cm$^{-3}$ $\mu$m$^{-1}$]); the Python map maker projects them into dust-band images (the IR is optically thin) |

| parameter | default | meaning |
|---|---|---|
| `sed_pah_live` | F | $x_{\rm HII}$-dependent PAH survival in the emission: the leaf spectrum is assembled from the model channels with the PAH channel weighted by $(x_{\rm HI} + f_{\rm ion,pah}\, x_{\rm HII})$ (needs a model with a 'PAH' channel, i.e. astrodust); turns the 8-$\mu$m map into the classic PAH ring at the ionization front |
| `dust_emis_transport` | T | transport the reradiated infrared through the same grain opacity, launching from each leaf in proportion to its emitted power with the wavelength drawn continuously from that leaf's spectrum; what the packets deposit returns to `Heat_dust`. Set F to post-process the emission only |
| `dust_niter`, `dust_tol` | 2, 1e-3 | maximum transport passes and the relative change in emitted luminosity that ends them; = 1 transports once with no self-consistency check |
| `dust_no_photons` | 1e6 | infrared packets per pass |

## 2.11 Peel-off imaging

| parameter | default | meaning |
|---|---|---|
| `ion_peel` | F | after convergence, one extra transport pass peels direct (stellar + diffuse) and dust-scattered contributions toward the observers → `_image` (blocks `direct_ion`/`scatt_ion`, plus `_fuv` with `add_fuv`) [erg s$^{-1}$ cm$^{-2}$ sr$^{-1}$] |
| `peel_bins` | F | bin-resolved cubes `direct_cube`/`scatt_cube` $(n_x, n_y, n_\nu)$ in addition to the band-integrated channel images (hardness maps, synthetic EUV/FUV photometry); the third (bin) axis is labeled by the `E_bin`/`dE_bin` [eV] and `wl_bin` [Å] arrays written once into the same file |
| `save_direc0` | F | also write the unattenuated direct image (`direc0_ion`/`_fuv`, and `direc0_cube` with `peel_bins`): the direct source with the path extinction $e^{-\tau}$ omitted, so `direc0`/`direct`= $e^{+\tau}$ is the line-of-sight optical depth toward the observer |

| parameter | default | meaning |
|---|---|---|
| obsx,obsy,obsz | unset | observer positions (or directions) in code units, up to 181; or use alpha/beta/gamma angle lists [deg] (phase_angle, inclination_angle, position_angle are accepted for −alpha, −beta, −gamma); default: one observer on $+z$ far away |
| distance | auto | observer distance (code units); unset, $10^3$ times the largest box half-extent (or the first obs radius when that is larger than ten half-extents) |
| nxim, nyim | 129, 129 | image pixels |
| dxim, dyim | auto | pixel scale [deg]; auto-sized to cover the box |

Peel-off also works with the external field: an external packet's direct peel uses the Lambert weight $\max(\hat{k}_{\rm obs}\cdot\hat{n}_{\rm in},0)/\pi$ about its entry-surface inward normal, so only the far-side surface contributes and the image is the background field seen through the medium (the cloud in silhouette); with save_direc0, direc0/direct $= e^{+\tau}$ is again the line-of-sight optical-depth map.

## 2.12 Output

out_file names the outputs: its directory part is dropped, a .h5, .hdf5, .fits or .fits.gz extension is stripped, and every file is <base>_<suffix> in the working directory. Left blank (the default) the base is the input file's name without .in. file_format = 'hdf5' (default; .h5; 'h5' is the same) or 'fits' (.fits.gz), in either case, sets the format of the section files (_rates, _emis, _heimeta, _dustemis, _image); the spectra and line tables are text.

## 2.13 Items carried over from MoCafe that MoCHII does not read

params_type (src/define.f90) still declares items of the MoCafe dust-continuum code that no MoCHII path reads, so that a MoCafe input still parses; an input that sets one of them to anything but its default is refused, with every such item named (setup_mod::refuse_unused_mocafe_items, which holds the list): the analytic density and source geometries (geometry, density_zscale, density_rscale, density_powerlaw_index, density_zprofile, spiral_m/pitch/amp, source_zscale, source_rscale, src_zscale, src_rscale, src_sersic_index, src_reff, src_axial_ratio, src_boxiness, src_spectrum, nr, cone_opening), the reflecting symmetries (xyz_symmetry, z_symmetry), the SED mode (use_sed, nlambda, lambda_min, lambda_max, lambda0, kext_file, source_spectrum, hgg, albedo, save_jlam, use_dustemis, dust_emission_method, dust_fast_table, dust_nU, dust_single_teq), and use_stokes, scatt_mat_file, use_reduced_wgt, use_mrw, mrw_gamma, allsky*, sightline_tau, use_amr_grid, output_normalization, radiation_angular_PDF_file, out_bitpix, out_bitpix_force. Thirty-three of these appear nowhere in src/ outside define.f90; the other nineteen are read only where they have no effect: the density geometries and use_sed by the box grid of grid_mod_car::grid_create, whose opacity grid_create_amr zeroes and whose optical depths it overwrites, the two symmetries by the walk after grid_create_amr has set them .false., and the SED-mode wavelength grid,

kext_file, source_spectrum, hgg and albedo by sed_mod::setup_sed, which nothing calls. Two MoCafe items with more than one value are kept with one: src_geometry must be 'point' in every slot, and amr_type must be 'generic' ('ramses' is refused; convert to the generic leaf file first). nphotons, nprint, nphotons_fuv, base_name, distance2cm, nobs, nscatt_tot and exetime are set by the code, and a value the input gives them is overwritten (not refused).

# 3. Output files

## 3.1 <base>_rates.h5

One dataset for each extension, leaf-ordered (.fits.gz with one HDU for each under file_format='fits'):

| extension | content |
|---|---|
| Gamma_HI/HeI/HeII | photoionization rates [$s^{-1}$] |
| Heat_HI/HeI/HeII | photoheating per particle [erg/s] |
| Heat_dust | grain heating [erg $s^{-1}$ $cm^{-3}$] (when ion_add_dust) |
| T_dust | equilibrium mixture dust temperature [K] (when ion_add_dust) |
| x_HI, x_HII, x_HeI, x_HeII, x_HeIII, n_e, T_e | converged state; every fraction to its own relative precision ($x_{\rm HI}+x_{\rm HII}=1$ and $x_{\rm HeI}+x_{\rm HeII}+x_{\rm HeIII}=1$ to rounding): beyond the ionization fronts $x_{\rm HII}$ and $x_{\rm HeIII}$ fall to $10^{-10}$–$10^{-100}$, where $1-x_{\rm HI}$ and $1-x_{\rm HeI}-x_{\rm HeII}$ keep none of their digits, so read x_HII and x_HeIII |
| ConvHist_* | stopping-test histories ConvHist_x_HII, ConvHist_HeI_HeII, ConvHist_HeII_HeIII (iteration, 2): the step $d$ and the signed step $s$ of each gas iteration, oldest first, for $x_{\rm HII}$, $He^0$–$He^+$ and $He^+$–$He^{2+}$ (the last only while $He^{2+}$ is present); restart_iteration_history reads them back |
| Phi_HeI_23S | He I $2^3S$ photoionization rate $\Phi_3$ [$s^{-1}$] of the last transport pass (with hei_metastable_coupling); restart_file reads it back, since the $2^3S$ split of the He I diffuse channel of the first pass is solved with it (a file without it restarts with $\Phi_3=0$, with a note) |
| x_<el>_stages | metal ion fractions (stage, leaf), with the keyword ABUND = $n$(X)/$n$(H) (with use_metals) |
| Gamma_<el>_stages, Heat_<el>_stages | metal photoionization rates [$s^{-1}$] and photoheating [erg $s^{-1}$ per ion] (transition, leaf) (with use_metals); restart_file reads the rates back |
| J_nu | band mean intensity (bin, leaf) [erg/s/$cm^2$/Hz/sr], with jnu_output |
| E_bin, dE_bin | band grid [eV] |
| LeafXYZ, LeafSize | leaf centers and full widths (code units; for slices/cutouts) |

The file header also carries the convergence record of the gas iteration: CONVERGD (did it converge), NITERDON (iterations run), FINALDX and FINALDTE (the cell maxima $\max|\Delta x_{\rm HII}|$ and $\max|\Delta T_e|/T_e$ of the final solve, or of the last iteration in a _rates_iterate file, whichever conv_crit is set), and STATESOL (the written state was solved in the written field: after the loop the last state is transported once more and the gas solved once more in that field, so the written

rates and the written state, metal stages included, belong together; false for rates-only runs). The rest of the header records the run: NNU_ION, NNU_BAND, EION_MIN, EION_MAX, TOT_LUM, DIST_CM, nphotons, with add_fuv EFUV_MIN, FUVSAMP and NPHOTFUV; the sampling (IONEMODE, CDFRTOL, NUBINUSE, ENRGSAMP, and QMCSEED under 'sobol'); and the ionizing-packet energy budget L_EMIT, L_ABS, L_ESC [erg s$^{-1}$].

## 3.2 Line and emissivity outputs

<base>_lines.txt is a text table of integrated line luminosities on the converged state, one row for each line: element, ionization stage, wavelength [Å], $L$ [erg s$^{-1}$], and $L/L(\mathrm{H}\beta)$. It is written by every run: the H and He recombination lines need only $n_e$, $T_e$ and the He ionization state, so an H/He model at fixed temperature gets its table too, with the metal blocks simply absent. With emis_output the same line set is also written leaf-by-leaf to <base>_emis (HDF5, or FITS with file_format='fits') as emissivity blocks in erg s$^{-1}$ cm$^{-3}$, where $\sum \mathrm{emis} \cdot V = L_{\mathrm{line}}$ and the block row order matches the text table. Every line comes from one of two sources: recombination-line tables (H and He, Table 14) and $n$-level statistical-equilibrium solves (the metals plus the optional collisional H I component, Table 15).

Table 14: Recombination-line blocks in <base>_emis; wavelengths in Å unless $\mu$m noted.

| block | source | lines | weighting, when present |
|---|---|---|---|
| emis_h | Storey & Hummer (1995) H I, case set by lyman_lines | H$\alpha$ 6565, H$\beta$ 4863, H$\gamma$ 4342, H$\delta$ 4103, Pa$\alpha$ 1.876 $\mu$m, Pa$\beta$ 1.282 $\mu$m, Br$\gamma$ 2.166 $\mu$m | always; $n_e\, n_{\mathrm{HII}}$ |
| emis_heii | SH95 He II (case per lyman_lines) | 1640, 3204, 4543, 4687, 5413, 10126 | only where an He III zone exists; $n_e\, n_{\mathrm{HeIII}}$ |
| emis_hei | Porter et al. (2012/2013) He I, case-B collisional-radiative (density-dependent) | 3890, 4027, 4473, 5877, 6680, 7067, 7283, 10833 | where He II is present; $n_e\, n_{\mathrm{HeII}}$ |
| emis_hc | 25-level H I collisional solve | H$\alpha$/H$\beta$/H$\gamma$ collisional component (separate hc rows) | only with h_coll_effects; $n_{\mathrm{HI}}$ |

The He I Porter table is a case-B collisional-radiative total (density-dependent; it already includes collisional excitation from the $2^3$S metastable), case B only; under case A a note is printed and the H I/He II tables switch to case A while He I stays case B.

The metal line blocks emis_<el>_<stage> follow the $n$-level solve. One block is written for every active registry ion (element abundance $> 0$) that has an atom in the selected model, data/atomic/excitation/<model>/nlevel_<el>_<stage>.txt (stage 1 is the neutral atom), with the lines of $L \geq$ emis_line_floor $L(\mathrm{H}\beta)$. Every metal abundance defaults to 0, so an element is active only when the input sets its par%abund_<El> $> 0$ (and use_metals, which then follows by default). Only the principal lines are listed below — the full transition list of each block is in

<base>_lines.txt. The list below dates from the first atoms (2026-07); with the full 'chianti' atoms every stage I–VII with an atom has a block.

Table 15: Metal collisional-line blocks (emis_<el>_<stage>); wavelengths in Å unless $\mu$m noted.

| ion (block) | principal lines |
|---|---|
| c_1 [C I] | 4623/4629, 8730, 9811, 9827, 9853, 370 $\mu$m, 609 $\mu$m |
| c_2 [C II] | 2325 (UV multiplet), 158 $\mu$m |
| c_3 C III] | 977, 1907, 1909, 178 $\mu$m |
| n_1 [N I] | 3468, 5199, 5202, 1.040/1.041 $\mu$m |
| n_2 [N II] | 3064/3071, 5756, 6550, 6585, 122 $\mu$m, 205 $\mu$m |
| n_3 [N III] | 1750 (UV multiplet), 57 $\mu$m |
| o_1 [O I] | 5579, 6302, 6366, 63 $\mu$m, 145 $\mu$m |
| o_2 [O II] | 2471, 3727, 3730, 7321/7331 |
| o_3 [O III] | 1661/1666, 2322/2332, 4364, 4960, 5008, 52 $\mu$m, 88 $\mu$m |
| ne_2 [Ne II] | 12.8 $\mu$m |
| ne_3 [Ne III] | 3343, 3870, 3969, 15.55 $\mu$m, 36 $\mu$m |
| s_2 [S II] | 4070/4078, 6718, 6733, 1.03 $\mu$m, 315 $\mu$m |
| s_3 [S III] | 3722, 6314, 8831, 9070, 9532, 12 $\mu$m, 18.7 $\mu$m, 33.5 $\mu$m |
| s_4 [S IV] | 1398–1424 (UV multiplet), 10.51 $\mu$m |
| *Elements outside the Lexington benchmark set* | |
| ar_3 [Ar III] | 3006/3110, 5193, 7138, 7753, 8.99 $\mu$m, 21.8 $\mu$m |
| ar_4 [Ar IV] | 2854/2869, 4713, 4741, 7170 (multiplet), 56 $\mu$m, 78 $\mu$m |
| mg_2 Mg II | 2796, 2804 |
| fe_2 [Fe II] | 1.26/1.32 $\mu$m, 5.34 $\mu$m, 17.9 $\mu$m, 26 $\mu$m (historical 40-line list) |
| fe_3 [Fe III] | 4659, 4703, 4881, 5011, 22.9 $\mu$m (historical 38-line list) |
| si_2 [Si II] | 2335 (multiplet), 34.8 $\mu$m |
| si_3 Si III] | 1206, 1883, 1892, 38 $\mu$m |
| si_4 Si IV | 1394, 1403 |
| cl_2 [Cl II] | 3587/3679, 6164, 8581, 9126, 14.4 $\mu$m, 33 $\mu$m |
| cl_3 [Cl III] | 3344/3354, 5519, 5539, 8480 (multiplet) |
| cl_4 [Cl IV] | 3120/3205, 5325, 7263, 7533, 11.8 $\mu$m, 20.3 $\mu$m |
| ca_2 Ca II | 3935, 3970, 7293/7326, 8500/8544/8665 |
| ca_5 [Ca V] | 3999, 5311, 6088, 3.05 $\mu$m, 4.16 $\mu$m, 11.5 $\mu$m |

Registry stages tracked in the ionization balance without an $n$-level file carry no line block: C IV, Ne I, Mg I, Mg III, Fe I, Fe IV, Si I, Si V, Ar I, Ar II, Cl I, Cl V, Ca I, Ca III, Ca IV. For Ar I–II, Fe I, Cl I, Cl V, Ca I and Ca III–IV CHIANTI v11.0.2 has no level or collision data; the others have CHIANTI data from which no $n$-level file has been made.

**State blocks.** Beyond the emissivities, <base>_emis carries the state a map needs without the rates file: LeafXYZ (leaf centers) with the DIST_CM keyword, LeafSize (leaf edge length; $V$ = (LeafSize·DIST_CM)$^3$), n_H, n_e, T_e, x_HI, x_HII, x_HeI, x_HeII, x_HeIII, and, with use_metals, the x_<el>_stages, Gamma_<el>_stages and Heat_<el>_stages blocks (same layout as the rates file). The Python reader EmisData, in tools/python/mochii_output.py, turns these blocks into line maps without the rates file.

## 3.3 Spectral and image outputs

| file | content |
|---|---|
| _nebcont.txt | (with solve_te) nebular continuum integrated over the grid: $\nu$ [Ryd], $\lambda$ [$\mu$m], total $L_\nu$ (free–bound + free–free + two-photon) [erg s$^{-1}$ Hz$^{-1}$], the free–free estimate below the H edge, and the two-photon part |
| _dustir.txt | (with ion_add_dust) equilibrium-mixture dust IR spectrum $L_\nu$ on 1–1000 $\mu$m with the energy-closure header |
| _dustsed.txt | (with ion_add_dust and dust_sed) SEDust stochastic dust SED with PAH bands |
| _dustesc.txt | (with ion_add_dust and dust_emis_transport) the infrared spectrum that actually escapes the grid, from the transported packets. Its ratio to the emitted spectrum in _dustir.txt/_dustsed.txt is the infrared escape fraction |
| _dustemis.h5 | (with dust_sed and dust_emis_bands) SEDust band emissivities of each leaf (emis_dust, wl_dust, LeafXYZ with DIST_CM, LeafSize), same block layout as _emis — readable by EmisData for dust-band maps |
| _heimeta.h5 | (with hei_metastable) the He I $2^3$S diagnostic: n_2s3 [cm$^{-3}$] (with the DIST_CM and A31 keywords), Phi_3 [s$^{-1}$], alpha_3 [cm$^3$ s$^{-1}$], Gamma_3 [erg s$^{-1}$ per atom], n_H, n_e, T_e, LeafXYZ, LeafSize — read by the Python HeiMetaData class for column and 10833 Å optical-depth maps, and convertible to a LaRT scattering-medium input by heimeta_to_lart.py |
| _image.h5 | (with ion_peel) peel-off images for each observer: direct_ion/scatt_ion (+_fuv) [erg s$^{-1}$ cm$^{-2}$ sr$^{-1}$], suffix _obs<k> when several observers; with peel_bins, direct_cube/scatt_cube ($n_x, n_y, n_\nu$) and the E_bin/dE_bin/wl_bin bin-grid arrays; with save_direc0, the unattenuated direc0_ion/_fuv (or direc0_cube) |
| _slab_Imu.txt | (with source_geometry = 'slab') the emergent intensity $I(\mu)$ at the two z-boundaries: columns $\mu$, $I_{\rm top}$, $I_{\rm bot}$ [erg s$^{-1}$ cm$^{-2}$ sr$^{-1}$] on slab_nmu bins ($F = 2\pi\int I\mu\,\mathrm{d}\mu$), with the reflected / transmitted / absorbed energy budget (fractions of the entering $L$) in the header |

## 3.4 Reading outputs in Python

`tools/python/mochii_output.py` is an importable module (works directly in a notebook) that reads every section file (HDF5 or FITS; `read_sections`), the line table (`read_lines_table`), and the text spectra (`read_spectrum`), and builds 2D maps from the emissivity file:

```
import sys; sys.path.insert(0, ".../MoCHII/tools/python")
import matplotlib.pyplot as plt
from mochii_output import EmisData

d = EmisData("run_emis.h5")
d.info()                            # blocks, fields, line count
d.line_list()                       # every stored line (block, wl)

# ready-made figures (notebook): returns (fig, ax)
fig, ax = d.plot_line("o_3", 5007.)                 # log S map
fig, ax = d.plot_line("h", 6563., unit="intensity",
                      photons=True)  # photons/s/cm^2/sr
fig, ax = d.plot_field("T_e", weight="EM")
fig, axs = plt.subplots(1, 2); d.plot_line("s_2", 6717., ax=axs[0]) \
    ; d.plot_line("s_2", 6731., ax=axs[1])   # doublet side by side

# raw arrays, when the figure is yours to build
img, ext = d.line_map("o_3", 5007., axis="z", npix=512)
img, ext = d.line_map("h", 6563., unit="intensity", photons=True)
img, ext = d.field_map("T_e", weight="EM")   # LOS-weighted average
Q = d.line_luminosity("h", 4861., photons=True)  # photon rate [1/s]
```

Line maps are flux-conserving: each leaf's luminosity is deposited with exact area overlap onto the pixel grid (AMR leaf sizes handled), so $\sum SA_{\rm pix} = L_{\rm line}$ to machine precision. `unit='flux'` (default) gives surface brightness [erg s$^{-1}$ cm$^{-2}$]; `unit='intensity'` gives specific intensity [erg s$^{-1}$ cm$^{-2}$ sr$^{-1}$] $= S/4\pi$ (isotropic emission, optically thin); `photons=True` converts either to photon units (divides by $hc/\lambda$ of the line). A `slab=(lo,hi)` cut restricts the line of sight. Field maps average `T_e`, `n_e`, `x_HI`, … along the line of sight with weight `'EM'` ($n_e^2 V$), `'ne'`, `'nH'`, or `'V'`. The same module doubles as a command-line tool for quick looks:

```
python3 mochii_output.py run_emis.h5                  # list lines
python3 mochii_output.py run_emis.h5 --line o_3 5007 \
        --npix 512 --log --out o3.png
python3 mochii_output.py run_emis.h5 --line h 6563 \
        --unit intensity --photons     # photons/s/cm^2/sr
python3 mochii_output.py run_emis.h5 --field T_e --weight EM
```

## 3.5 Slices and cutouts

A *slice* is the actual leaf value on a plane (the cell that contains each pixel), distinct from the line-of-sight *projection* above — a slice resolves the ionization front sharply where a projection blurs it. `tools/python/leaf_field.py` holds the backend (`leaf_slice`, the nearest-leaf fallback `leaf_slice_nn`, and `plot_slice`); `mochii_output.py` exposes it on the output files:

```
from mochii_output import EmisData, RatesData
r = RatesData("run_rates.h5")                # leaf state
fig, ax = r.plot_field_slice("T_e", axis="z", coord=0.0)
fig, ax = r.plot_field_slice("x_HI", axis="z", coord=0.0, log=True)
img, ext = r.slice_field("n_e", axis="x", coord=0.0,
                         bounds=(-2,2,-2,2))  # cutout window


d = EmisData("run_emis.h5")
fig, ax = d.plot_line_slice("o_3", 5007., axis="z")   # emissivity slice
```

Exact slices need the leaf full widths: the `_emis.h5` file always carries `LeafSize`, and `_rates.h5` does too; a rates file written before `LeafSize` existed falls back to a nearest-leaf slice (a one-time note is printed). The command line takes `--slice FIELD --axis z --coord 0` on either file.

## 3.6 He I $2^3$S metastable and 10833 Å absorption

The `_heimeta` file (`hei_metastable`) is read by the `HeiMetaData` class, which adds a line-of-sight column map and the 10833 Å line-center optical depth:

```
from mochii_output import HeiMetaData
d = HeiMetaData("run_heimeta.h5")
d.info()                                  # n_2s3, Phi_3, line constants
img, ext = d.column_map("n_2s3")          # LOS column [cm^-2]
img, ext = d.tau10833_map(v_turb=0.0,     # tau_0 map (km/s turbulence)
                          component="doublet")  # blended J'=1,2 pair
fig, ax = d.plot_column("n_2s3", log=True)
fig, ax = d.plot_tau10833(component="all")
```

`column_map` is the same flux-conserving deposit as the line maps with weight $nV$, so $\sum NA_{\rm pix} = \sum nV$ to machine precision. `tau10833_map` deposits the line-center opacity $\kappa_0 = \frac{\sqrt{\pi}e^2}{m_e c} f \frac{\lambda}{b} n_3$, $b = \sqrt{2kT_e/m_{\rm He} + v_{\rm turb}^2}$, so a pixel value is $\int \kappa_0 \, {\rm d}l$. `component` selects the resolved $J' = 0$ line (10832.06 Å), the blended $J' = 1,2$ `'doublet'` (10833.22/10833.31 Å, 0.089 Å apart, inside the thermal Doppler width), or `'all'`; the constants ($\lambda$, $f$, $A$) come from `data/atomic/hei_metastable.txt`. The map warns when line-center $\tau_0 > 1$: the diagnostic assumes optically thin 10833 (line transfer is out of scope). The command line takes `--column n_2s3` or `--tau10833 --component doublet --vturb 5`.

Where line transfer *is* wanted (resonant-scattering emission profiles from the metastable medium), `heimeta_to_lart.py` (in `tools/python/`) converts a `_heimeta` file into a scattering-medium input for the resonance-line code LaRT (line `HeI_10833`): `--format car` (default, uniform grids) writes the all-in-one Cartesian file with `gasDen` = $n_3$, `T` = $T_e$, and zero velocities; `--format amr` writes the generic AMR table with $n_3$ in the `n_ion` column. `--field` selects a different leaf field for the scatterer. A worked end-to-end case (converted grid + LaRT namelist) is in `examples/hei_metastable/lart/`.

### 3.7 Building AMR grids

`tools/python/AMR_grid/` builds the generic AMR octree file the code reads (`grid_type='amr'`): `AMR_grid.py` (the `AMRGrid` class — uniform, density-gradient, or resolution refinement; `set_density`/`set_neutral_fraction`; `write` to HDF5/FITS/`.fits.gz`/text with columns `x,y,z,level,nH[,metallicity,xHI,ndust]`) and `make_amr_sphere.py` (a sphere/shell CLI). `AMRGrid` carries the same `slice`/`plot_slice`/`cutout` so a grid can be inspected before a run.

## 4. Worked examples (the regression tests)

Each `tests/` directory is self-contained: grid builder or a symlinked grid, the input files, and a gate: a `check_*.py` script, or for the data-level gates a `check_*.f90` program built and run by `run_check.sh` (a few use `run_test.sh`, `build_test.sh`, `check_io.sh` or `compare_state.sh`). `tests/README.md` describes each gate in full. The stored outputs under `tests/` are dated records, not before/after references for a code change.

| directory | what it demonstrates / gate |
|---|---|
| `g0_gamma` | rate integrals vs. analytic attenuation (bias +0.004%) |
| `g1_stromgren` | Strömgren sphere, case B fixed $T_e$; AMR $\equiv$ uniform |
| `g2a_thermal` | thermal H/He nebula vs. the 1D exact reference |
| `g2_hii` | MOCASSIN Lexington HII20/HII40 benchmarks + line fluxes |
| `g4_refine` | I-front re-refinement vs. native level 7 |
| `g4_tng` | Illustris-TNG post-processing demo (shadows, maps) |
| `d_dusty` | dusty Strömgren; scattering bracket; $T_d$/IR |
| `peel` | peel-off images vs. the optically thin analytic flux (+0.017%); bin cubes |
| `uni_dda` | uniform grid: `'amr'` walk $\equiv$ octree (bit-identical); DDA walk to rounding |
| `pdr` | FUV/PDR switches: carbon-electron shell, photoelectric $T_e$ |
| `ext_field` | isotropic external field: interior $J = J_{\rm ext}$ to 0.2% (rec and sph) |
| `multi_src` | multiple point sources: superposition, spectrum resolution, one-file spectra |
| `peel_ext` | external-field peel: silhouette flux vs. $JA_{\rm proj}/d^2$, chord $\tau$ spectrum |
| `qmc` | quasi-random launch: Sobol generator vs. `scipy` + Owen-net balance; the fixed-state variance gate (cell-wise $\Gamma_{\rm HI}$ gain 29$\times$ at $2^{17}$, 222$\times$ at $2^{20}$) |
| `hei_meta` | He I $2^3$S metastable: analytic limits of the balance, module $n_3$ vs. the Python evaluation of the data file on each run's state, $\Phi_3$ against an independent $J_\nu$ integration, and the 10833 Å column/optical-depth diagnostic (conservation + hand integral) |

| directory | what it demonstrates / gate |
|---|---|
| `hei_metastable_coupling` | the $2^3$S coupling: Bray et al. (2000) Tables 2/3 at their nodes, $\alpha_3 = f_{3S}\alpha_B$, closed-form $n_3$ vs. a $2 \times 2$ solve, energy-ledger closure, the diffuse low-density limit, and `solve_ion_cell` balances with the $2^3$S terms in partially neutral gas |
| `photo_shells` | metal photoionization cross section and photoelectron heating of all 66 metal transitions against Verner's `phfit2.f` on 5–1000 eV (outer-shell fit below the first inner edge, subshell sum above; the Ne III table at its nodes and its join) |
| `ne3_ground_term_photoionization` | the Ne III `PHTABLE` against the three NORAD level tables: the cross section, the channel-6 $\alpha_1$ by the Milne relation, the sampled spectrum across the resonances |
| `metal_recombination_continuum` | diffuse channel 6 on all 66 transitions: the outer-shell cross section against `phfit2.f`, the band coefficient against quadrature, energy bookkeeping, the packet photon rate, the hydrogenic C VI check |
| `freefree_emission` | diffuse channel 7 and the free–free cooling: the frequency integral of the spectral coefficient against the cooling ($Z$ = 1–6), the band coefficient, the packet photon rate |
| `freefree_absorption` | diagnostic: the share free–free absorption would have in the band opacity of a run (an upper bound) |
| `grain_photoelectric` | WD01 grain charging, photoelectric heating and accretion cooling (their Tables 4 and 5, eqs. 44–45), the charge distribution against Draine's charging routine, the HM79 exchange, the cost of the 4.78 eV band cut |
| `ion_stages` | stages I–VII: the stage solvers, the registry, rates, inputs, the thermal and charge-exchange consumers and every excitation atom of both models, linked against the production objects (`run_checks.py`) |
| `fitting` | unit tests of the offline generators under `tools/fitting/` |
| `hcoll` | atomic-hydrogen impact on the C I, C II, O I ground terms: low-density photon rates vs. the published tables, Boltzmann populations at high $n_{\rm HI}$, thermal increment = emissivity difference |
| `charge_exchange` | metal–H charge exchange conserved in each reacting pair, the hydrogen root solve, detailed balance; He–H charge exchange (rate forms, balances, charge neutrality); charge-exchange heating (a defect for every carried reaction, defects against the ionization potentials and product terms, detailed-balance pairs cancel) |
| `hei_on_the_spot` | case-B on-the-spot He I photons: the ground-capture share $y$ against a direct quadrature, the excited exits against the cascade, the heat assembled independently, `solve_ion_cell` balances with the sources, and the Osterbrock & Ferland $p \simeq 0.96$ limit (0.966) |
| `heii_on_the_spot` | case-B on-the-spot He II photons: the shares against a direct quadrature, the parts against the channel-5 luminosities, the limits with no $H^0$ or no $He^0$, the heat, the balances, and (g) the absorption limit (He/H = 0.2: $x_{\rm HI} \approx 0$, absorbed fraction $< 1$, balance closed with it; He/H = 0.1: fraction 1) |

| directory | what it demonstrates / gate |
|---|---|
| `fuv_stratified` | band-stratified FUV launch: the launched luminosity of each stratum, the field estimators of `'separate'` against `'proportional'` and the run without the extension over four seeds, and the Sobol ionizing stratum identical to the run without it |
| `slab` | plane-parallel slab: beam transmission $e^{-\tau}$, oblique incidence, DDA against `'amr'` walk, tile-size independence, energy closure |
| `continuous_energy` | continuous packet energies: thresholds, Planck and table transport, metal coupling, bin-count, rank-count and random/Sobol independence |
| `energy_sampler` | the source samplers: tabulated and Planck inverse CDFs against analytic references, random/Sobol statistics |
| `convergence` | the `conv_crit = 'vol'` stopping test on synthetic histories |
| `restart` | `restart_file`: same-grid restart bit for bit, a different grid by position, files without `x_HII`/`x_HeIII` |
| `grain_kext` | the served `/kext` curve against the size integral of the built grain model, for the four models |
| `milne` | the Milne ground-continuum sampler and $\alpha_1(T)$ against quadrature and Cloudy c25.00 |
| `hydrogenic_recombination` | $\alpha_B$, $\alpha_A$ and the $2s$ shares of the H I/He II cascade against SH95 and Osterbrock & Ferland |
| `hei_cascade` | He I two-photon sampler, branch fractions, 584 Å split |
| `heii_cascade` | the He III cascade channel: $n = 2$ branching, path shares, samplers, case A |
| `secondary_ionization` | fast-photoelectron partition against Shull & van Steenberg (1985) and Dalgarno et al. (1999) |
| `cooling_ne` | the $(T, n_e)$ line-cooling table against the direct $n$-level solve |
| `rates` | `tools/python/mochii_rates.py` against the Fortran rates |
| `io` | HDF5 error propagation, HDF5-off build, extension detection |
| `thermal_parallel` | serial against MPI-divided thermal solve, state compared exactly |
| `cloudy_c25` | not a gate: Cloudy c25.00 runs of the Lexington benchmarks and the fixed-state cooling comparison |

A minimal thermal + metals + dust input on the `d_dusty` geometry (the gate itself runs fixed-$T_e$ and gas-only, with the switches of its day pinned; this input takes the current defaults, among them the transported diffuse continuum, and names the metals it carries):

```
&parameters
 par%no_photons = 4.0e7          par%iseed = 1234
 par%grid_type = 'amr'           par%amr_file = 'amr_sphere_L6_nH100.fits'
 par%distance_unit = 'pc'        par%dust_density_law = 'global_dgr'
 par%nnu_ion = 32
 par%eion_min = 13.598           par%eion_max = 100.0
```

```
 par%tstar = 4.0e4              par%luminosity = 3.178e38
 par%xHI_init = 1.0e-3          par%abund_He = 0.1
 par%gas_niter = 60             par%conv_crit = 'vol'
 par%gas_tol = 5.0e-3           par%gas_tol_te = 1.0e-2
 par%abund_C = 2.2e-4           par%abund_N = 4.0e-5
 par%abund_O = 3.3e-4           par%abund_Ne = 5.0e-5
 par%abund_S = 9.0e-6
 par%ion_add_dust = .true.
 par%out_file = 'run.h5'
/
```

`solve_te` is on by default, `use_metals` follows from the abundances, `cext_dust` is taken from the default astrodust grains, and `data_dir`/`atomic_dir` resolve next to the executable.

# 5. Adding an ion

Adding an ion touches only data:

1. The atom and its cooling fit: for the `'chianti'` model the full CHIANTI atom is written by `tools/fitting/chianti_cooling.py` (every bound collision-connected level and every row) and its low-density cooling fitted by `tools/fitting/fit_cooling.py`, into `data/atomic/excitation/chianti/` with the `manifest.json` updated (`tools/fitting/chianti_excitation_model_coverage.py`; checks `validate_chianti_excitation_model.py`; `md/atomic_stage_expansion/chianti_model_2026-10-11/`). Check the reported maximum fit error. `fit_nlevel.py` writes the older Chebyshev form, which the reader still accepts.

2. `tools/fitting/make_element_data.py`: add the element to `ELEMENTS` (Z, stage count) and its thresholds; charge exchange entries where sources exist; run. The CHIANTI `.rrparams`/`.drparams` types 1–3 are handled (`RR`/`RR2`/`DR`/`DR2` lines). The photoionization rows are parsed from Verner's `phfit2.f` itself and follow its prescription: `PHOTO` $E_{\rm th}\,E_0\,\sigma_0\,y_a\,P\,y_w\,y_0\,y_1$ (the VFKY96 outer-shell fit, used below the first inner-shell edge; written when that edge is above zero), `PHINNER` $E_{\rm inn}$ (`phfit2`'s `einn`; 0 for the elements absent from the VFKY96 table, e.g. Cl, which then use the subshell sum at every energy), and one `PHSHELL` $l\,E_{\rm th}\,E_0\,\sigma_0\,y_a\,P\,y_w$ row for each occupied subshell (Verner & Yakovlev 1995), summed at and above $E_{\rm inn}$. The reader refuses a transition with no `PHSHELL` rows, or with $E_{\rm inn} > 0$ and no `PHOTO` row. A `PHTABLE` $n$ record followed by $n$ rows $\epsilon\ \sigma$ (photoelectron energy $\epsilon = E - E_{\rm th}$ [eV], cross section [Mb], linear in $\epsilon$) replaces `PHOTO` below its last node, where it must end on the `PHOTO` fit; only Ne III carries one (Nahar 2019 R-matrix, written by `tools/fitting/ground_term_photoionization_table.py`), and the reader refuses a table that is short, not increasing from $\epsilon = 0$, negative, reaches $E_{\rm inn}$ or does not join the fit. A charge-exchange rate given as a table is `CXR 8` followed by `CXRTAB` $n\ T_1..T_n\ k_1..k_n$ ($n \le 1024$; K and cm$^3$ s$^{-1}$, interpolated log–log, $T$ held at the table ends), as for the Barragan et al. (2006) $N^{2+}$ and $O^{2+}$ rows. A transition resolved by the product term of the ion is `CXI 9` and `CXR 9` followed by `CXSTATE` $n_{\rm term}\ n_T\ g_{\rm react}\ \Delta$IP, `CXSTATET` $T_1..T_{n_T}$ and one `CXSTATEK` *label* $l_{\rm lo}\ l_{\rm hi}\ g\ E\ k_1..k_{n_T}$ row

for each term (its levels in `nlevel_<el>_<stage+1>.txt`, weight, energy [eV], forward rate into it): the forward rate is the sum, the reverse each channel's detailed balance weighted by the term's population from that $n$-level atom at the local $(T, n_e)$ (sulfur, Zhao et al. 2005).

3. Atomic-hydrogen impact (optional, for a neutral-gas coolant): a `hcoll_<el>_<stage>.txt` beside the $n$-level file in `data/atomic/excitation/<model>/`, written by `tools/fitting/make_hcoll_fine_structure.py`: a `NHCOL` $n$ line, then $n$ rows `lo hi form Tmin Tmax c_1..c_10` with the level indices of the $n$-level file; form 1 is the Abrahamsson et al. (2007) Eq. 1 excitation fit ($c_1 = b$, $c_{2\ldots} = a_i$), form 2 a de-excitation power law $c_1(T/100)^{c_2}$. It is picked up automatically and used under `neutral_hydrogen_excitation` (default on).

4. Fortran (only for a NEW element; new stages of a known element need nothing): a `par%abund_<El>` component of `params_type` in `define.f90`, one entry in the name/abundance lists of `species_mod::species_setup`, and one in the abundance list of `setup.f90::read_parameters_with_implicit_defaults` (`NAB`, `ABNAME`, `set_abundances`, `abundances`), which decides `use_metals` from the abundances an input gives.

5. Verify: extend `verify_nlevel_pyneb.py` with a diagnostic ratio where PyNeb has data; run a smoke test and check the new lines in `_lines.txt`.

Data caveats met so far: Ar II, Ca IV and Cl VII have no CHIANTI 11 valence collision data (Stout atoms are used); Ar I, Cl I, Ca I, Fe I, Cl V and Ca III have no atom (stages tracked without lines); Fe III collision strengths differ from PyNeb's BB14 by up to ∼16% in 4658/4702 (known spread for iron); charge-exchange stage labels differ between the Huang (2023) and MOCASSIN transcriptions for Mg/Fe (the Huang labels are adopted; see `make_element_data.py`).

# 6. Practical notes

- **Bins**: the photoionization and heating rates come from path estimators at the exact packet energies, so the bin count (`nnu_ion`, `nnu_fuv`) sets only the resolution of the diagnostic spectra (`J_nu`, the peel-off cubes, the SEDust leaf spectrum), never the solution. The `add_fuv` extension carries its own bins below `eion_min`.

- **FUV and the PDR**: without `add_fuv` there is no radiation beyond the I-front and the metals there recombine toward neutral; with it, FUV penetrates the I-front and photoionizes the metal stages whose thresholds lie below 13.6 eV (C I 11.3, S I 10.4, Si I 8.2, Fe I 7.9, Mg I 7.6 eV, . . . ), producing the PDR-side C II and Mg II. In the FUV band the gas opacity is exactly this low-threshold metal photoionization (`ion_metal_abs`, on by default; H and He do not absorb below 13.6 eV), so the FUV is attenuated by that photoionization together with dust — only where those metals are already ionized does dust extinction alone remain. The metal stage fractions iterate with this opacity self-consistently, and the corresponding metal photoheating is in the thermal balance (`metal_heat`, on by default); with dust the WD01 grain photoelectric heating (`grain_pe`) and the gas–grain exchange are on by default, and without $H_2$/CO chemistry PDR-zone temperatures are first-order estimates.

- **Convergence**: use `conv_crit = 'vol'` for production — the cell-max criterion stalls at the front-cell noise floor (and, with `add_fuv`, at no-heating cells where $T_e$ pins to `te_min`). Both measures are printed each iteration. The explicit diffuse field smooths the front and converges far better.

- **Starts**: prefer an ionized initial state; a neutral start advances the front ∼one cell per iteration.
- **kext tables**: the default astrodust and DL07 tables cover the EUV directly (astrodust: $1.0 \times 10^{-4}$–$3.981 \times 10^{4}\,\mu$m; DL07: $6.2054 \times 10^{-5}$–$3.981 \times 10^{4}\,\mu$m). If `ion_dust_kext` names an explicit table, that table must cover the transported band; tables are sorted on read (the D03 file has an out-of-order seam).
- **Re-refinement**: one event per run; the old shared-memory windows are reclaimed only at exit.
- **SEDust**: the grain-model data are resolved from `sed_data_dir` (blank: `<executable dir>/SEDust/data`). Building the grain model is a startup cost every rank pays (measured 2026-08-07 at `sed_NT` = 200: 153 s astrodust, 2.4 s DL07, 0.18 s Zubko) whenever the run carries dust; the exact stochastic solve then costs ∼0.05–0.1 s for each gas leaf (distributed over ranks; not re-measured since).